Dec 1, 2025

Selected Topics in Computational Quantum Physics

量子物理计算方法选讲

Shuo Yang（杨硕）

Department of Physics, Tsinghua University

Email: shuoyang@tsinghua.edu.cn

WeChat: condmat-ys

Tensor networks

- Projected Entangled-Pair States (PEPS) and topological order
- 2D tensor network algorithms
- Multi-scale Entanglement Renormalization Ansatz (MERA)

Selected review articles:
F. Verstraete, J.I. Cirac, and V. Murg, arXiv: 0907.2796.
R. Orus, arXiv:1306.2164.
N. Schuch, arXiv:1306.5551.
J. C. Bridgeman and C. T. Chubb, arXiv:1603.03039.
R. Orus, arXiv: 1812.04011.
J.I. Cirac, D. Perez-Garcia, N. Schuch, F. Verstraete, arXiv:2011.12127.

New progress (update daily)
http://quattro.phys.sci.kobe-u.ac.jp/dmrg/condmat.html

Methods to obtain 2D PEPS states

- imaginary time evolution

$$e^{-\beta H}\,|\psi\rangle = \sum_i e^{-\beta E_i}\,|\phi_i\rangle \overset{\beta\to\infty}{\longrightarrow} e^{-\beta E_0}\,|\phi_0\rangle$$

(1) simple update: cheap and simple, not accurate

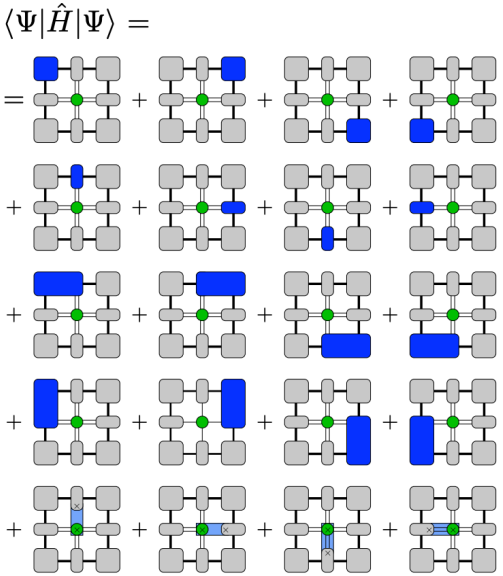
(2) full update: more accurate, more expensive

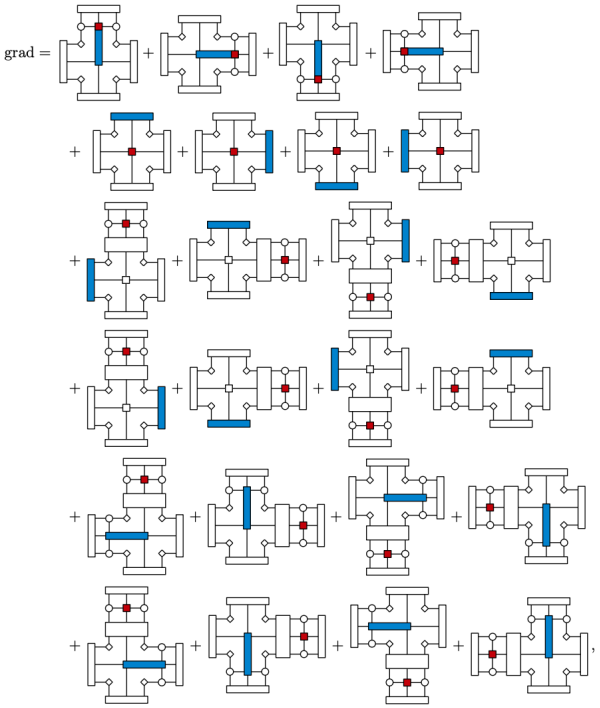
- energy minimization

find a PEPS which minimize the energy

$$E = \frac{\langle\psi|\,H\,|\psi\rangle}{\langle\psi\,|\psi\rangle}$$

even more accurate, but more complicated



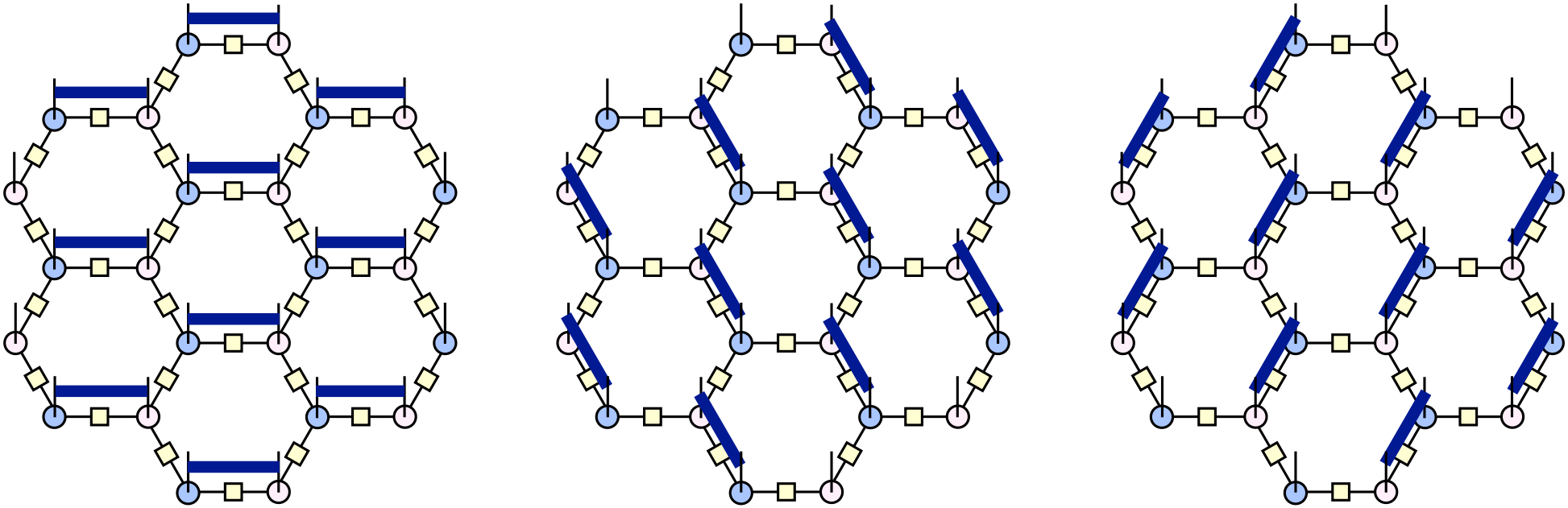


L. Vanderstraeten, J. Haegeman, P. Corboz, F. Verstraete, PRB 94, 155123 (2016)
P. Corboz, PRB 94, 035133 (2016)

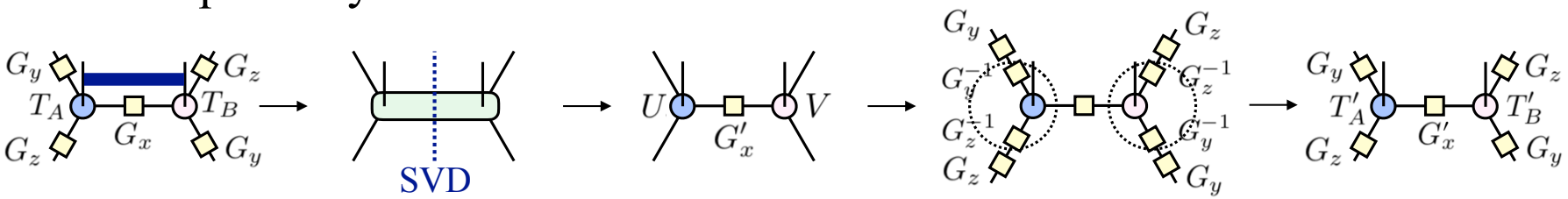
Simple update

- apply the imaginary time evolution operators in all directions in turn

$$H = H_x + H_y + H_z \qquad e^{-\tau H} \approx e^{-\tau H_x} e^{-\tau H_y} e^{-\tau H_z} + O\left(\tau^2\right)$$



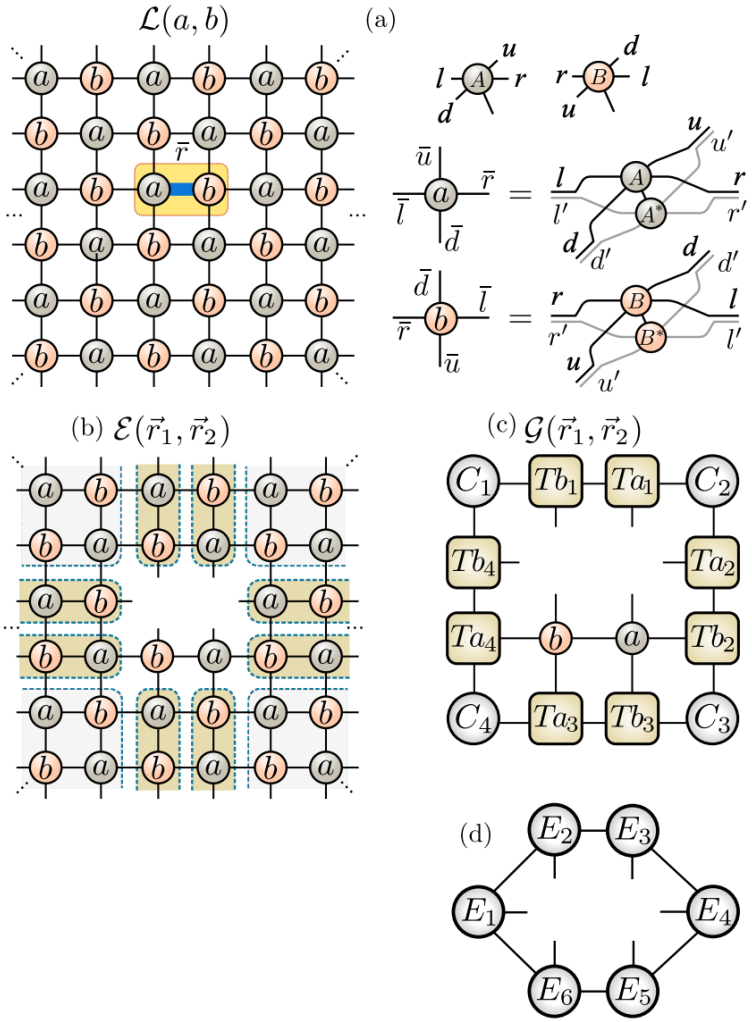
- local update by SVD

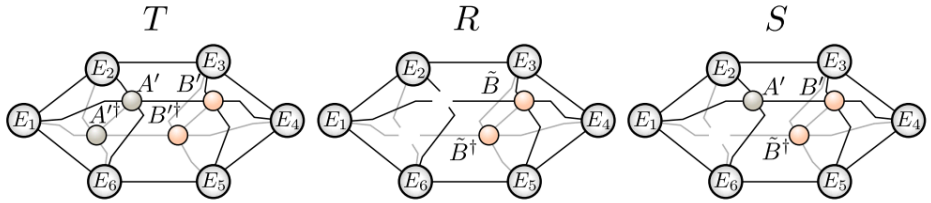


H. C. Jiang, Z. Y. Weng, and T. Xiang, PRL 101, 090603 (2008)
Z.-C. Gu, M. Levin, and X.-G. Wen, PRB 78, 205116 (2008)

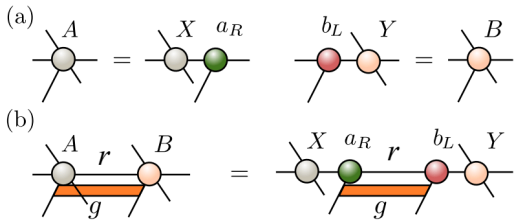
Full update

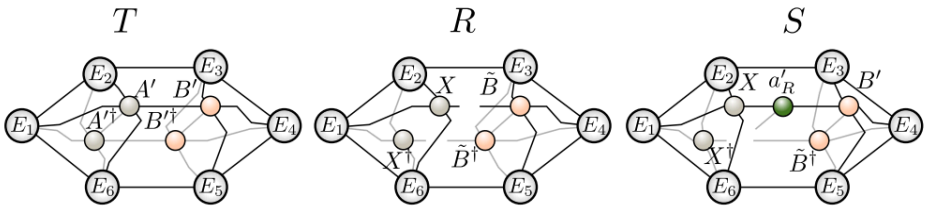
- take environment into account





$$d(\tilde{A},\tilde{A}^\dagger) = \tilde{A}^\dagger R\tilde{A} - \tilde{A}^\dagger S - S^\dagger\tilde{A} + T$$





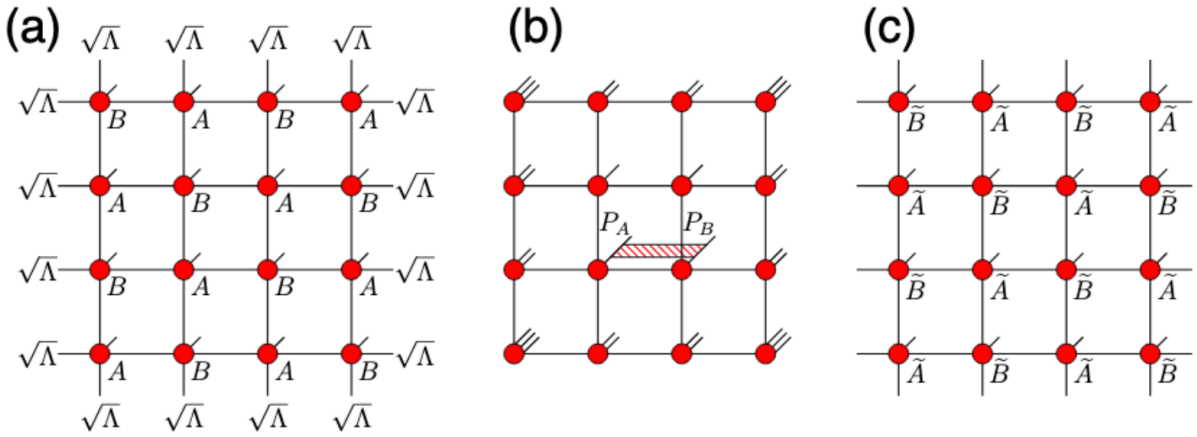
$$d(\tilde{a}_R,\tilde{b}_L) = \langle\Psi_{a'_R b'_L}|\Psi_{a'_R b'_L}\rangle + \langle\Psi_{\tilde{a}_R\tilde{b}_L}|\Psi_{\tilde{a}_R\tilde{b}_R}\rangle - \\ -\langle\Psi_{\tilde{a}_R\tilde{b}_L}|\Psi_{a'_R b'_L}\rangle - \langle\Psi_{\tilde{a}_R\tilde{b}_L}|\Psi_{a'_R b'_L}\rangle.$$

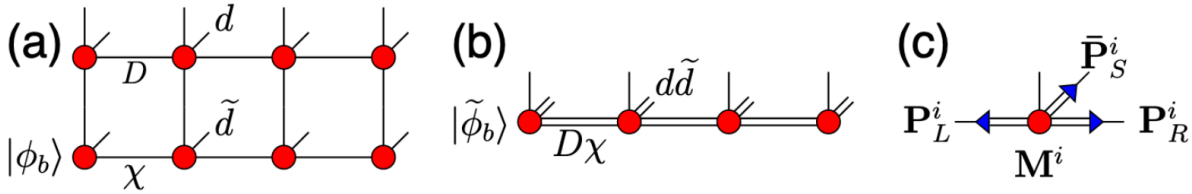
$$d(\tilde{a}_R,\tilde{a}_R^\dagger) = \tilde{a}_R^\dagger R\tilde{a}_R - \tilde{a}_R^\dagger S - S^\dagger\tilde{a}_R + T$$

J. Jordan, R. Orus, G. Vidal, F. Verstraete, and J. I. Cirac, PRL 101, 250602 (2008)
M. Lubasch, I. Cirac, and M.C. Banuls, PRB 90, 064425 (2014)
H. N. Phien, J. A. Bengua, H. D. Tuan, P. Corboz, and R. Orus, PRB 92, 035142 (2015)

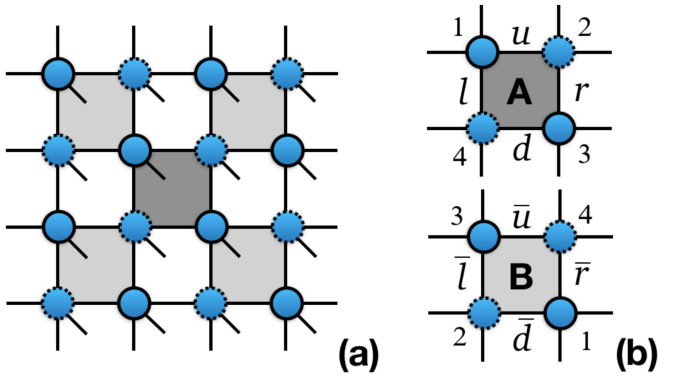
Cluster update & Loop update

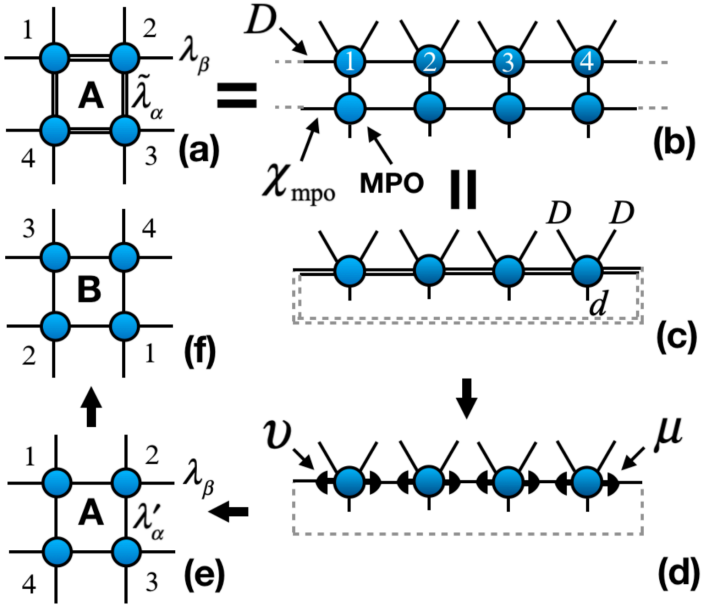
- Cluster update





- Loop update

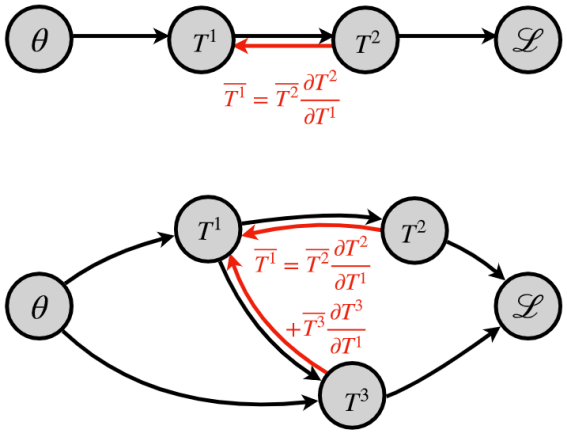


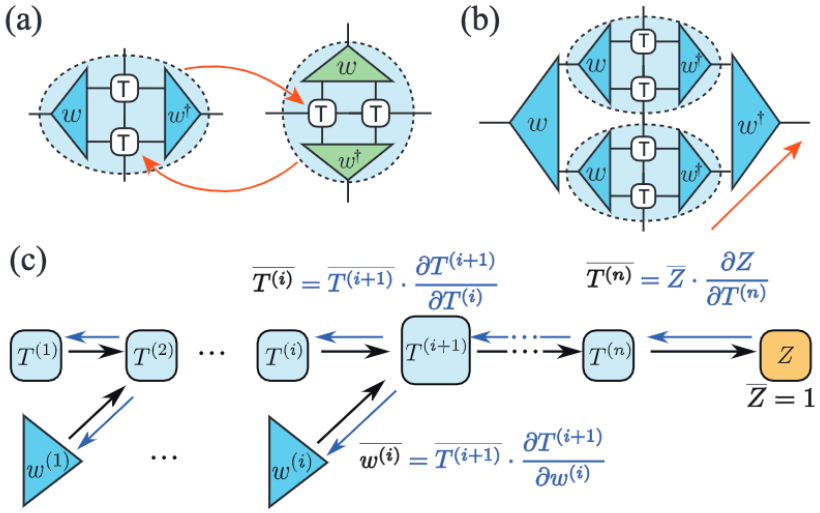


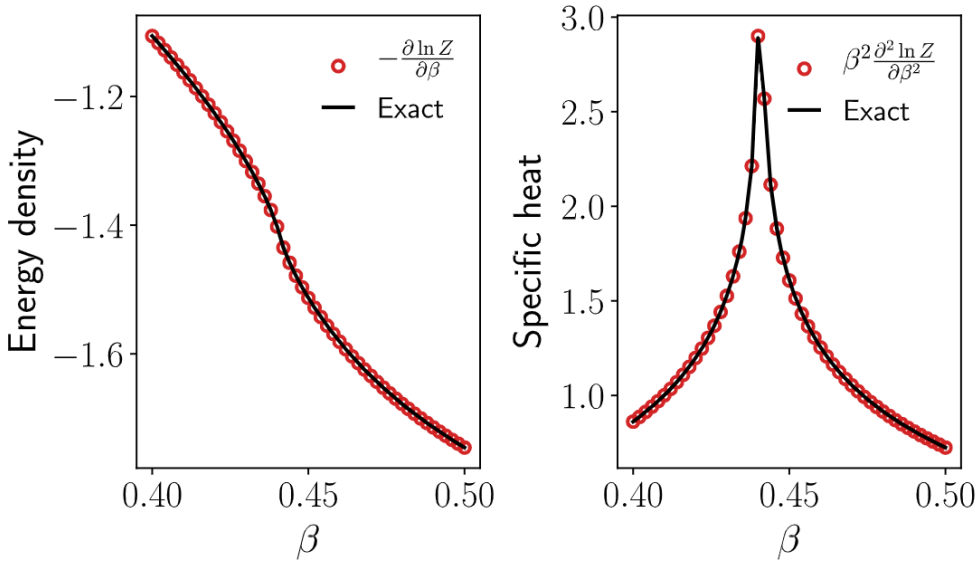
L. Wang and F. Verstraete, arXiv: 1110.4362

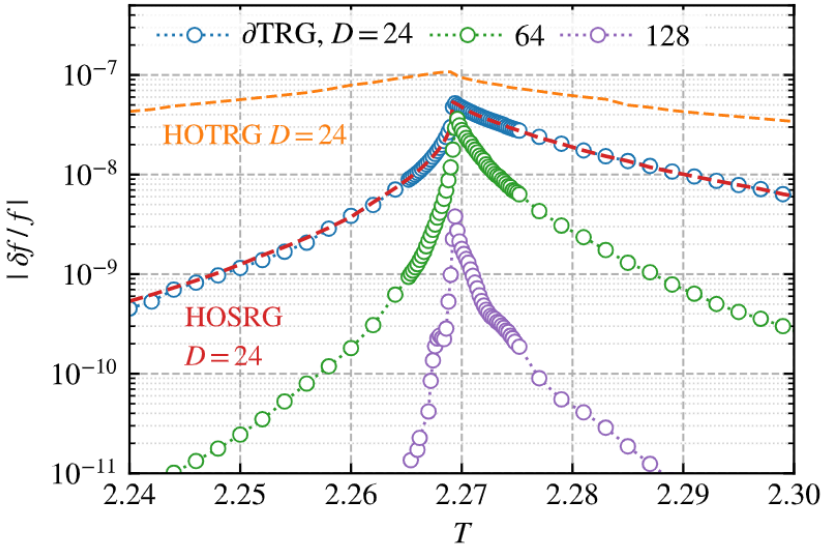
Y. Zheng and S. Yang, PRB 102, 075147 (2020)

Differentiable Programming Tensor Networks









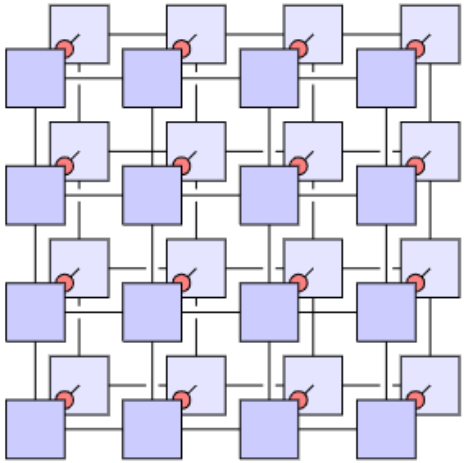
Hai-Jun Liao, *et. al*, PRX 9, 031041 (2019)

Bin-Bin Chen, *et. al*, PRB 101, 220409 (2020)

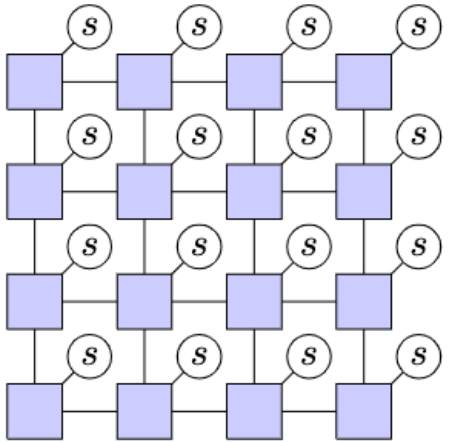
Monte Carlo sampling of PEPS

$$\langle \Psi(\{A\})|\Psi(\{A\})\rangle = \sum_{\{\mathcal{S}\}} |\langle \mathcal{S}|\Psi(\{A\})\rangle|^2$$

$$\mathbb{1} = \quad = \sum_{\{s\}}$$

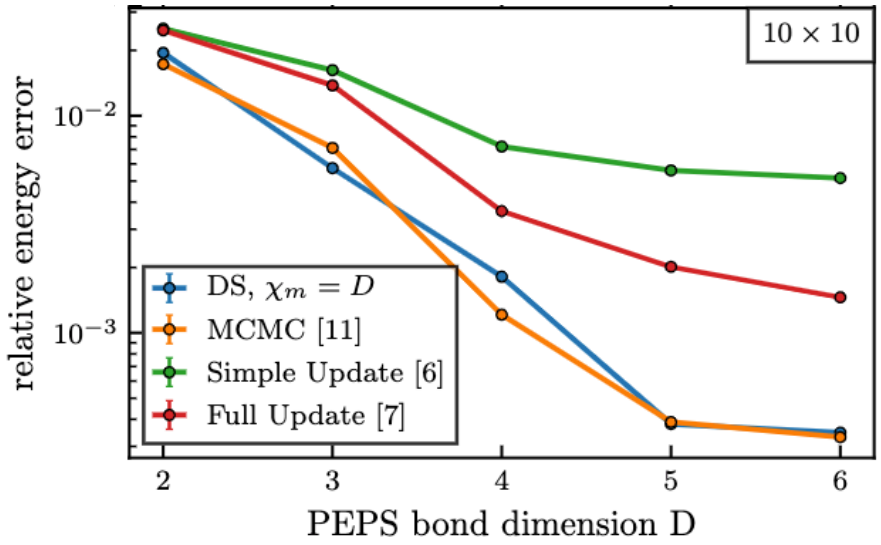


$$\Psi(\mathcal{S}) =$$



$$\langle \widehat{O} \rangle = \frac{\langle \Psi|\widehat{O}|\Psi\rangle}{\langle \Psi|\Psi\rangle} = \sum_{\{\mathcal{S}\}} \frac{|\Psi(\mathcal{S})|^2}{\langle \Psi|\Psi\rangle} \frac{\langle \mathcal{S}|\widehat{O}|\Psi\rangle}{\Psi(\mathcal{S})} = \sum_{\{\mathcal{S}\}} \frac{|\Psi(\mathcal{S})|^2}{\langle \Psi|\Psi\rangle} O_{\mathrm{loc}}(\mathcal{S}),$$

$$O_{\mathrm{loc}}(\mathcal{S}) = \langle \mathcal{S}|\widehat{O}|\Psi\rangle / \Psi(\mathcal{S}), \qquad p_\Psi(\mathcal{S}) \equiv \frac{|\Psi(\mathcal{S})|^2}{\langle \Psi|\Psi\rangle}$$



L. Wang, I. Pizorn, and F. Verstraete, PRB 83, 134421(2011)

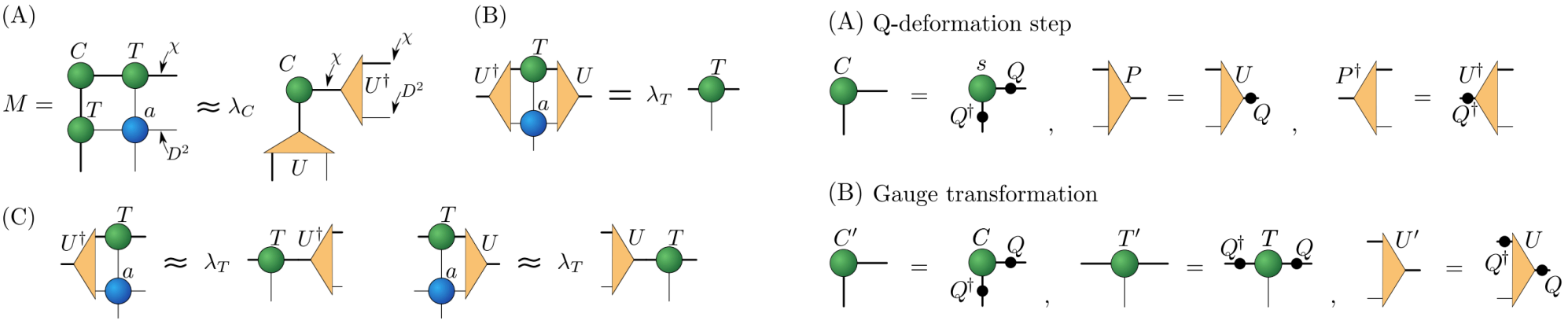
T. Vieijra, J. Haegeman, F. Verstraete, L. Vanderstraeten, arXiv:2109.07356

Stable and efficient differentiation of tensor network algorithms

Anna Francuz,[1,*] Norbert Schuch,[1,2] and Bram Vanhecke[1,†]

[1] *University of Vienna, Faculty of Physics, Boltzmanngasse 5, 1090 Wien, Austria*
[2] *University of Vienna, Faculty of Mathematics, Oskar-Morgenstern-Platz 1, 1090 Wien, Austria*

Gradient based optimization methods are the established state-of-the-art paradigm to study strongly entangled quantum systems in two dimensions with Projected Entangled Pair States. However, the key ingredient, the gradient itself, has proven challenging to calculate accurately and reliably in the case of a corner transfer matrix (CTM)-based approach. Automatic differentiation (AD), which is the best known tool for calculating the gradient, still suffers some crucial shortcomings. Some of these are known, like the problem of excessive memory usage and the divergences which may arise when differentiating a singular value decomposition (SVD). Importantly, we also find that there is a fundamental inaccuracy in the currently used backpropagation of SVD that had not been noted before. In this paper, we describe all these problems and provide them with compact and easy to implement solutions. We analyse the impact of these changes and find that the last problem – the use of the correct gradient – is by far the dominant one and thus should be considered a crucial patch to any AD application that makes use of an SVD for truncation.



Anna Francuz, Norbert Schuch, and Bram Vanhecke, arXiv:2311.11894

Final project rules

- please find your own project, which can be anything related to this course
  please submit the source code and a detailed note to explain your code and results
  please prepare a PPT for the presentation (no more than 2 minutes)
  please submit all the above through http://learn.tsinghua.edu.cn
  please also submit your PPT file to Tsinghua Cloud, file name is your name
  https://cloud.tsinghua.edu.cn/u/d/883ff777aac841fe9037/

- deadline is Dec. 29, 6:00 pm
  if you submit it after the deadline, you will have fewer points
  the final deadline for all previous homework is Dec. 31, 23:59 pm

- final presentation time: Dec. 29, during class
  the detailed schedule will be announced later

- please let me know if you have any questions

Grading of the final project

- academic integrity [+30]

  completing assignments independently, creating and expressing your own ideas, DON'T copy answers from others or allow others to copy your answers

- the source code can be executed and can provide correct results [+10]

  the source code has high readability [+10]

- there is a detailed note file to explain the basic principle, source code and results [+10]

  the note file is well-written and easy to understand [+10]

- there is a PPT for presentation [+10]

  the presentation is clear and interesting [+10]

- difficulty-dependent score: easy: [+0], medium: [+10], hard: [+20]

  time-dependent score: after DDL: [-1 per day]

  self-motivation score: new ideas or extra results [+10]

Final project samples

- calculate spectral function using the exact diagonalization method
- simulate the time evolution of a 1D quantum system using the TDVP method
- calculate the dispersion relation of a 1D quantum system using the tangent-space MPS method
- simulate fermionic 1D systems using the variational MPS method
- simulate 2D quantum models using Quantum Monte Carlo methods
- contract 2D tensor networks using the boundary MPS method
- read a relevant research paper and reproduce some results

...

Tensor networks

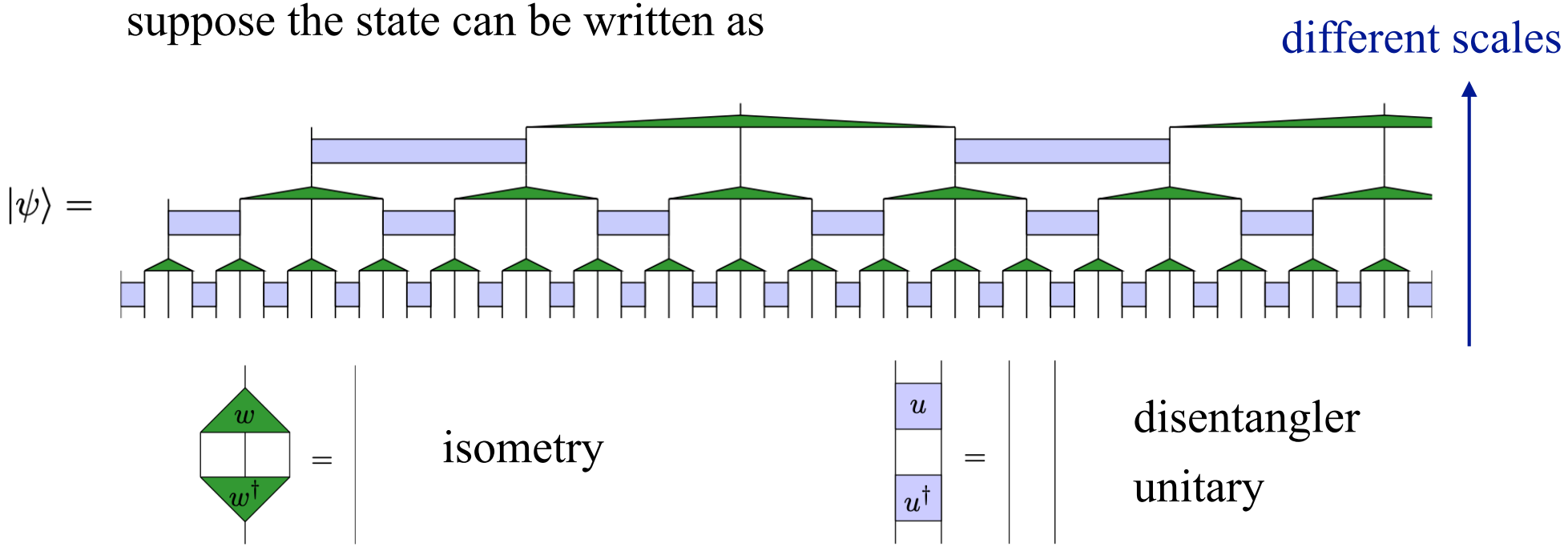
- Projected Entangled-Pair States (PEPS) and topological order
- 2D tensor network algorithms
- Multi-scale Entanglement Renormalization Ansatz (MERA)

G. Vidal, Phys. Rev. Lett. 99, 220405 (2007)

Multi-scale Entanglement Renormalization Ansatz (MERA)

- well suited to represent ground states of gapless Hamiltonians in 1D

  suppose the state can be written as



- how many layers? $\log_3 N$ layers

- u and w tensors do not have to be identical

  if they are identical, we expect translationally and scale invariant states

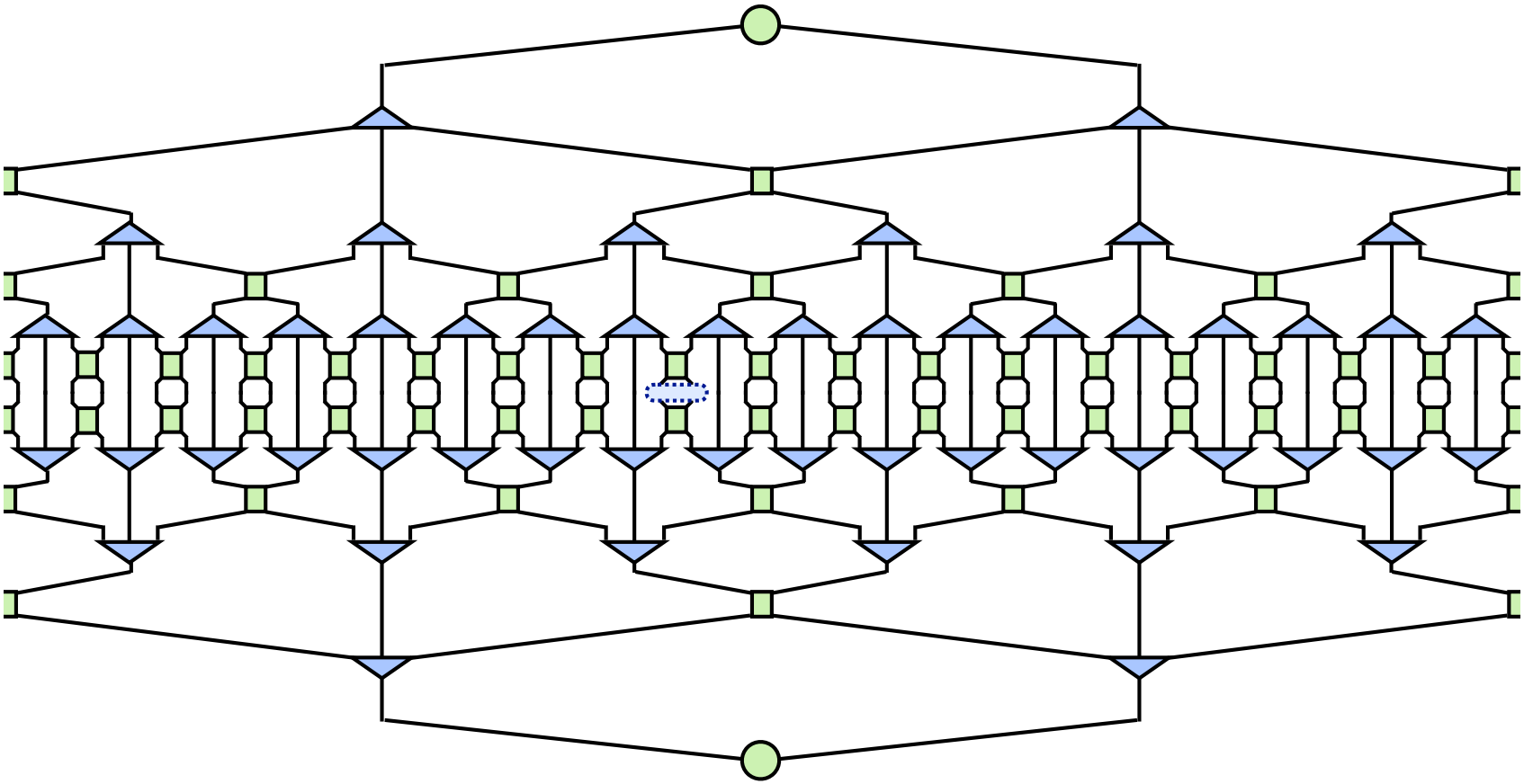
- can be straightforwardly generalized to higher dimensional systems

Contraction of MERA

- isometric and unitary constraints are necessary in order to contract MERA efficiently

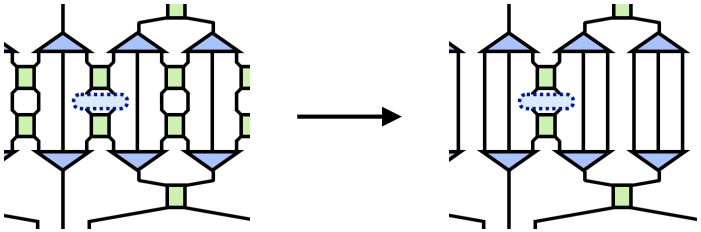
- compute the reduced density matrix for sites (r_1, r_2)

$$\rho(r_1, r_2) = \mathrm{Tr}_{\overline{(r_1, r_2)}} \left| \Psi \right\rangle \left\langle \Psi \right|$$

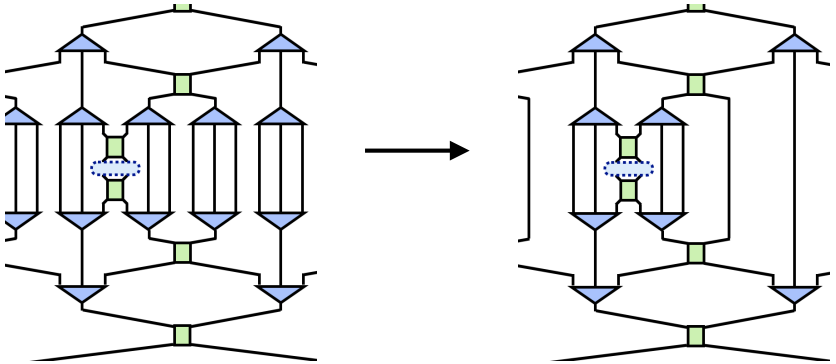


Contraction of MERA

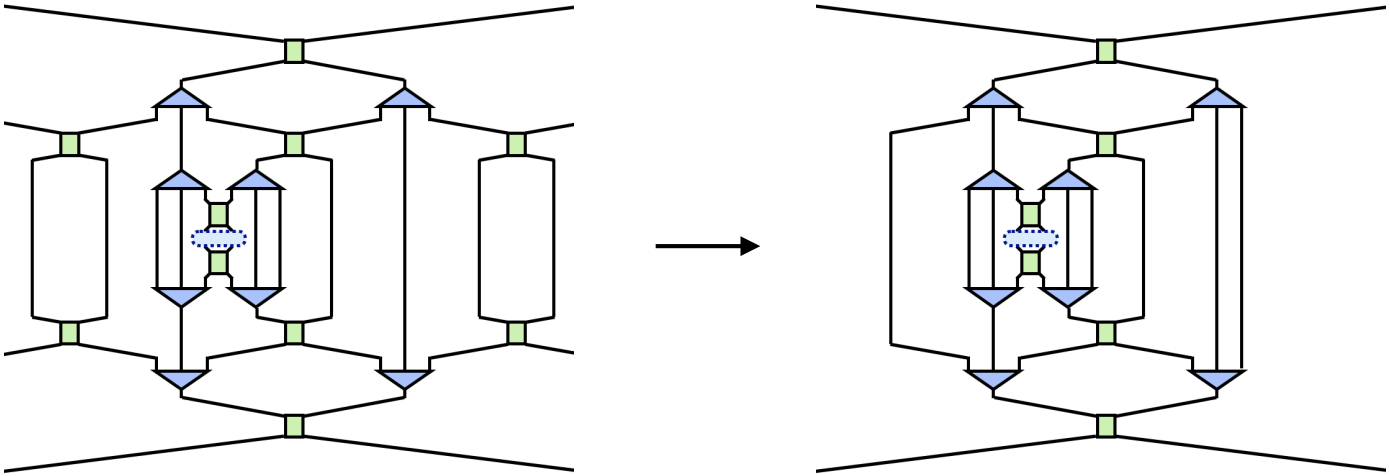
- use unitary



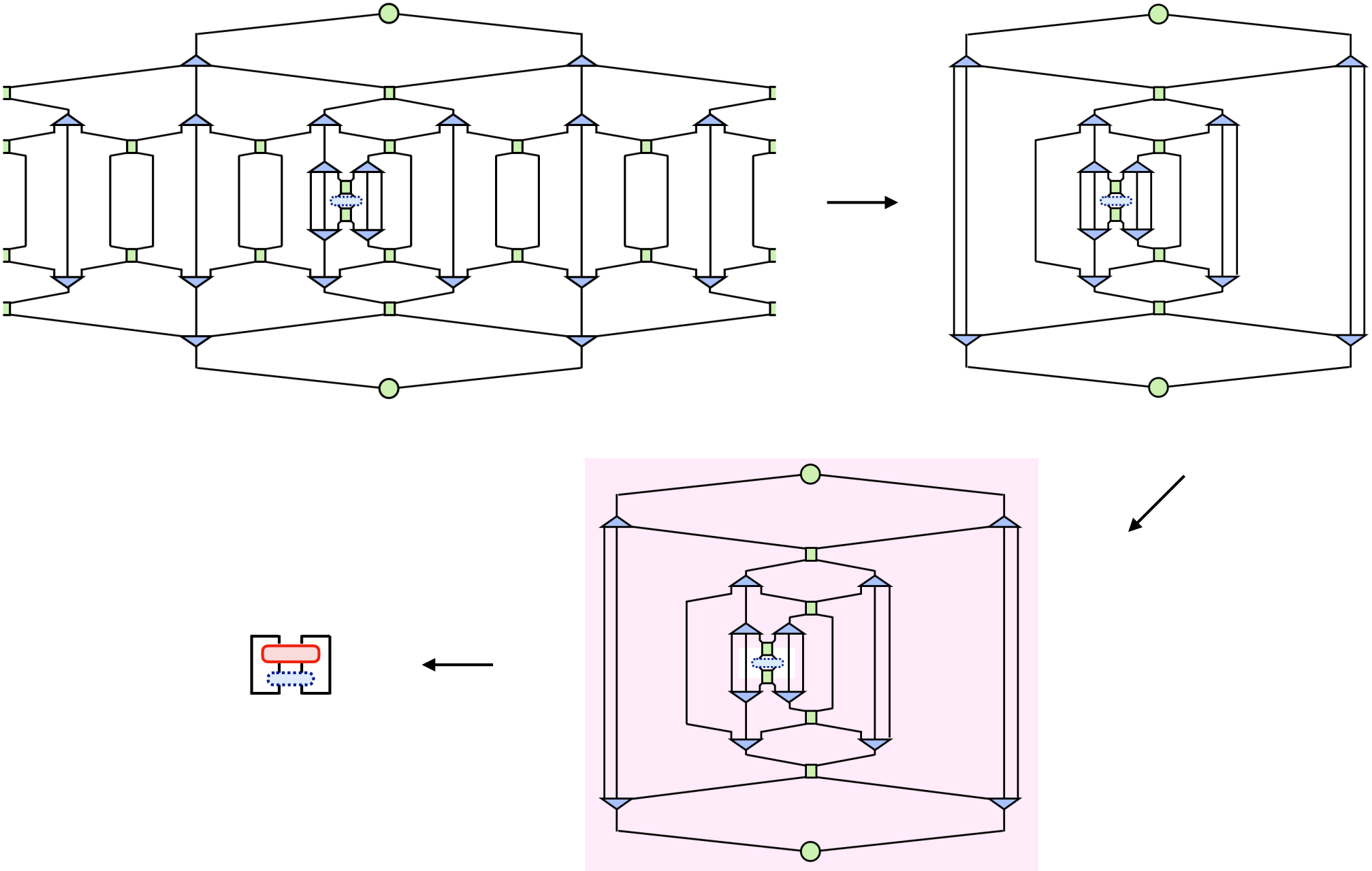
- use isometry



- use unitary

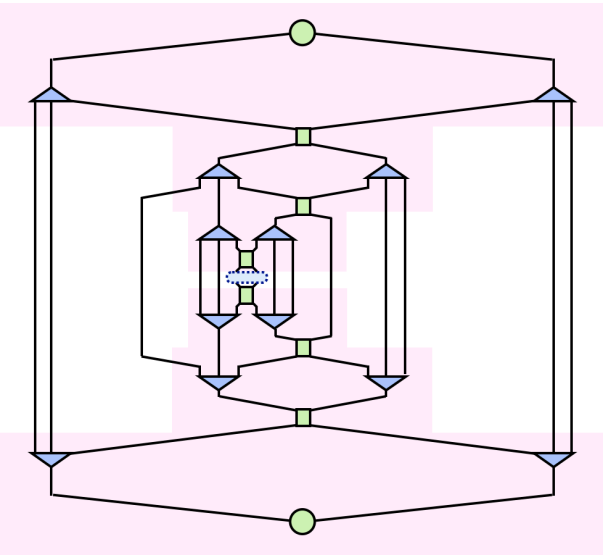


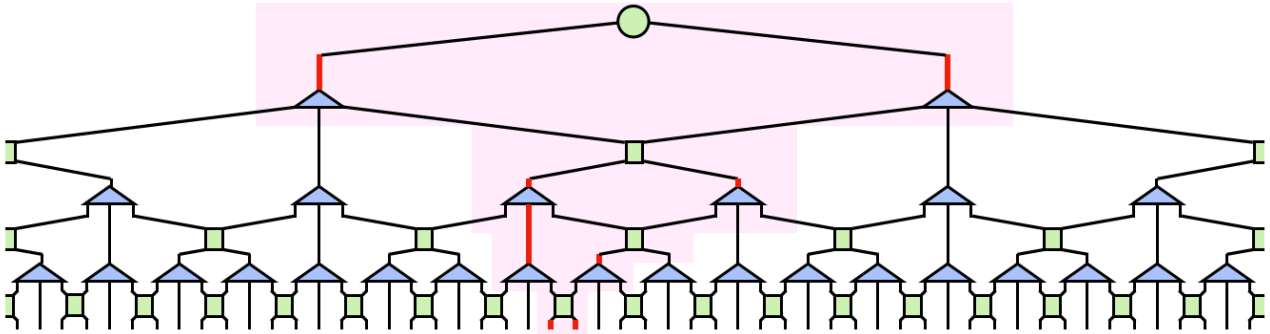
Contraction of MERA



Causal cone

- only a few of the tensors from the MERA contribute to the density matrix





- the causal cone of site (r_1, r_2) is the set of tensors that affect the reduced density matrix $\rho\ (r_1, r_2)$

- the causal cone of any pair of neighboring sites involves at most two sites at every level of MERA → bounded causal width

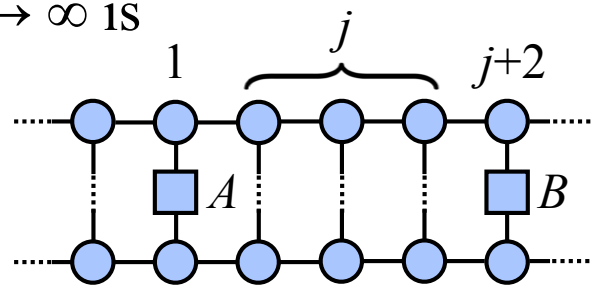
- contraction cost is proportional to number of layers $\log_3 N$ → efficient manipulation

Exponential decay of Correlations in MPS

- we consider a uniform MPS

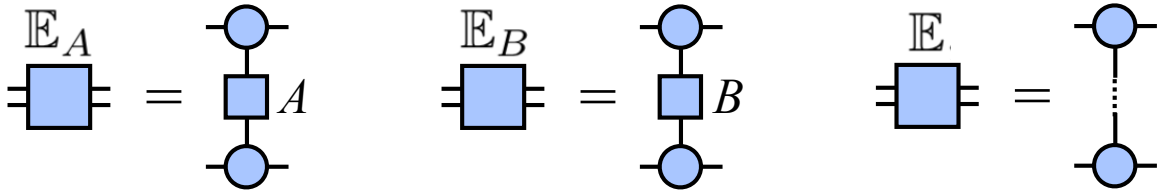
  the correlation function between operator A at site 1 and B at site j+2 on a chain of length $N \to \infty$ is

$$\langle A_1 B_{j+2} \rangle =$$



$$= \frac{\mathrm{Tr}\left[\mathbb{E}_A \mathbb{E}^j \mathbb{E}_B \mathbb{E}^{N-j-2}\right]}{\mathrm{Tr}\left[\mathbb{E}^N\right]}$$

  where



- we assume $\mathbb{E}$ has a unique maximal eigenvalues λ_1, which we normalize to $\lambda_1 = 1$

  $\mathbb{E}\,|r\rangle = |r\rangle$, $\langle l|\,\mathbb{E} = \langle l|$, for large N, $\mathbb{E}^N \to |r\rangle\langle l|$

  we now expand $$\mathbb{E}^j = |r\rangle\langle l| + \sum_{k\geq 2} \lambda_k^j\, |r_k\rangle\langle l_k|$$

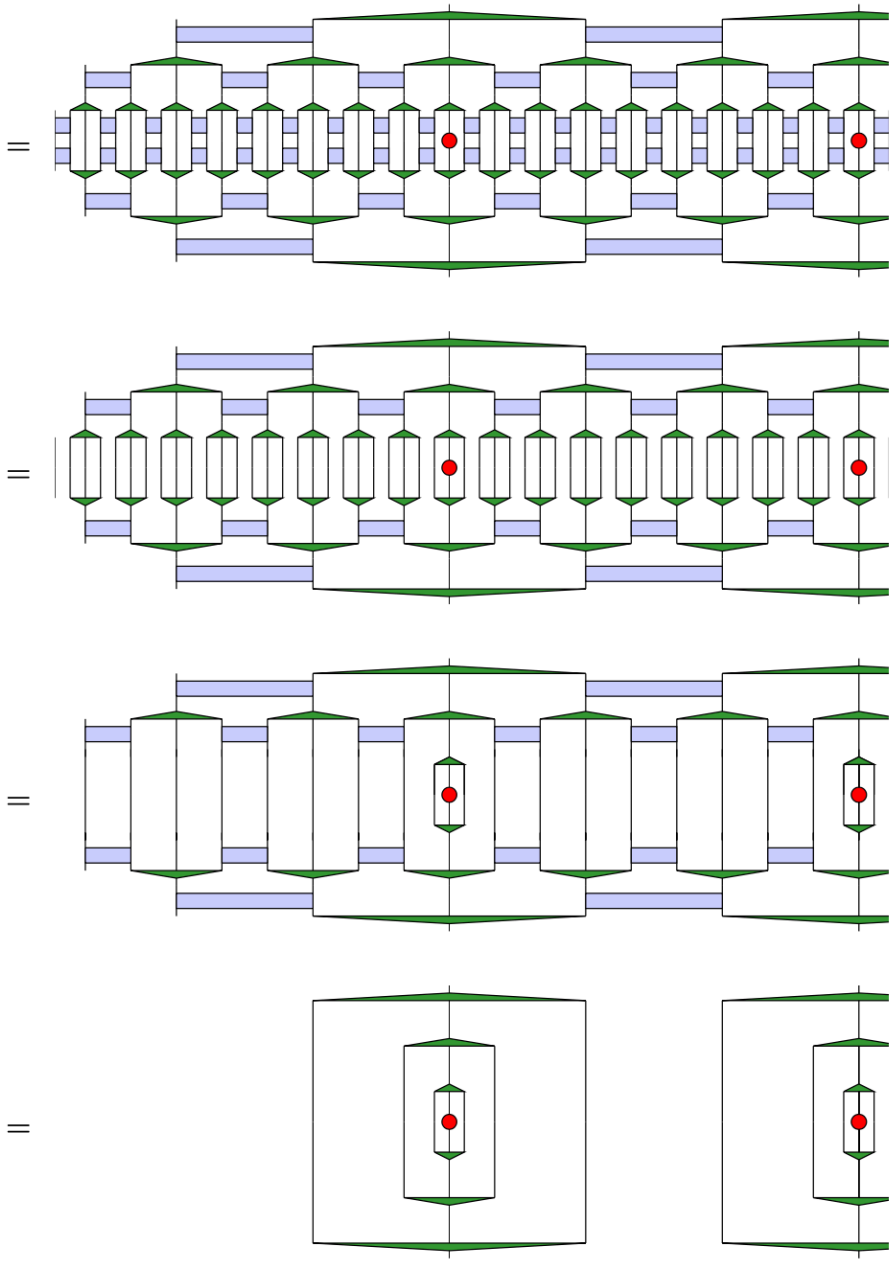
  we find that $$\langle A_1 B_{j+2} \rangle = \langle l|\,\mathbb{E}_A\,|r\rangle\langle l|\,\mathbb{E}_B\,|r\rangle + \sum_{k\geq 2} \lambda_k^j\, \langle l|\,\mathbb{E}_A\,|r_k\rangle\langle l_k|\,\mathbb{E}_B\,|r\rangle$$

  and thus decays exponentially with j, $\langle A_1 B_{j+2} \rangle = e^{-j/\xi}$

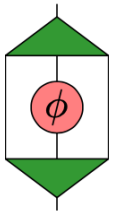
  the correlation length is given by $\xi = -1/\ln|\lambda_2|$

Power law decay of Correlations in MERA

- two-point correlator $\langle\psi| O_j O_{j+r} |\psi\rangle$



- the length scale behavior of the correlator is completely determined by the application of a superoperator

$$\mathcal{S}(\phi) =$$  $$S(\phi) = \lambda\phi$$

- two-point correlator

$$\langle O(0)O(r)\rangle = \mathrm{Tr}(\rho S^{\log_3 r}(O) S^{\log_3 r}(O))$$

$$\approx \lambda^{2\log_3 r} = \frac{1}{r^{2\Delta}}$$

power law decay of correlators

scaling dimensions $\Delta = -\log_3 \lambda$

- this is because

$$2\log_3\lambda \log_3 r = 2\log_3 r \log_3 \lambda$$

$$\lambda^{2\log_3 r} = r^{2\log_3 \lambda}$$

Scaling of entanglement entropy

- key property realized in MERA which cannot be realized in MPS

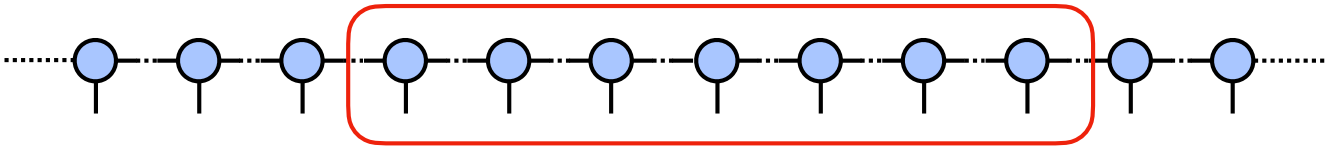
- can be easily seen by bond counting

  if n bonds must be broken to separate a region from the rest of the network

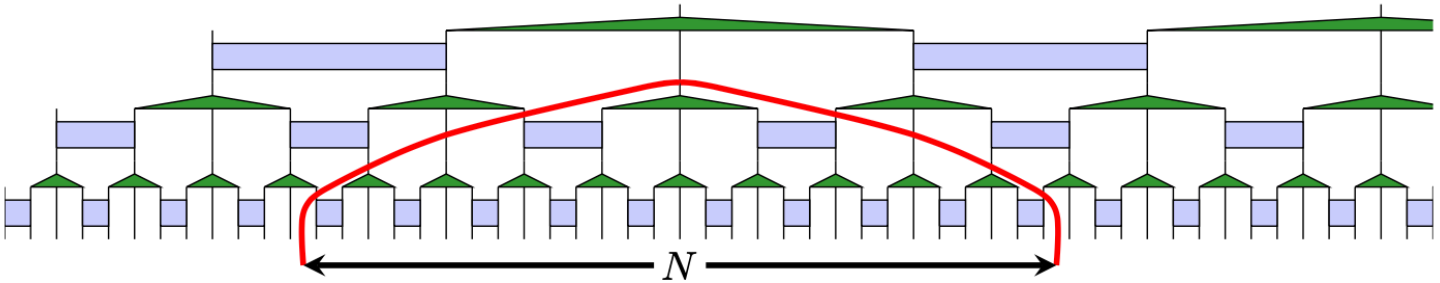
  the maximum entanglement of entropy that can be supported is $n\log D$

  D is the bond dimension

- MPS: cutting $n = 2$ bonds, maximum entanglement entropy is a constant

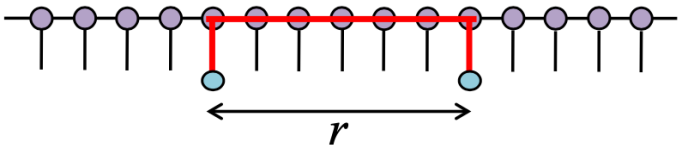


- MERA: cutting $n \sim O(\log_3 N)$ bonds, maximum entanglement entropy scales as $\log_3 N$



Tensor networks for 1D lattices

- Matrix Product State (MPS)

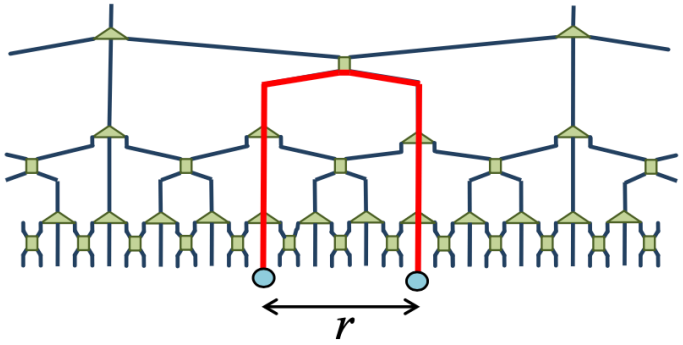


- 1D non-critical phases (gapped Hamiltonian)
- saturation of entanglement entropy: $S <$ const
- exponential decay of correlators

$$\langle O(0)O(r)\rangle \sim e^{-r/\xi}$$

- easy to implement

- Multi-scale Entanglement Renormalization Ansatz (MERA)



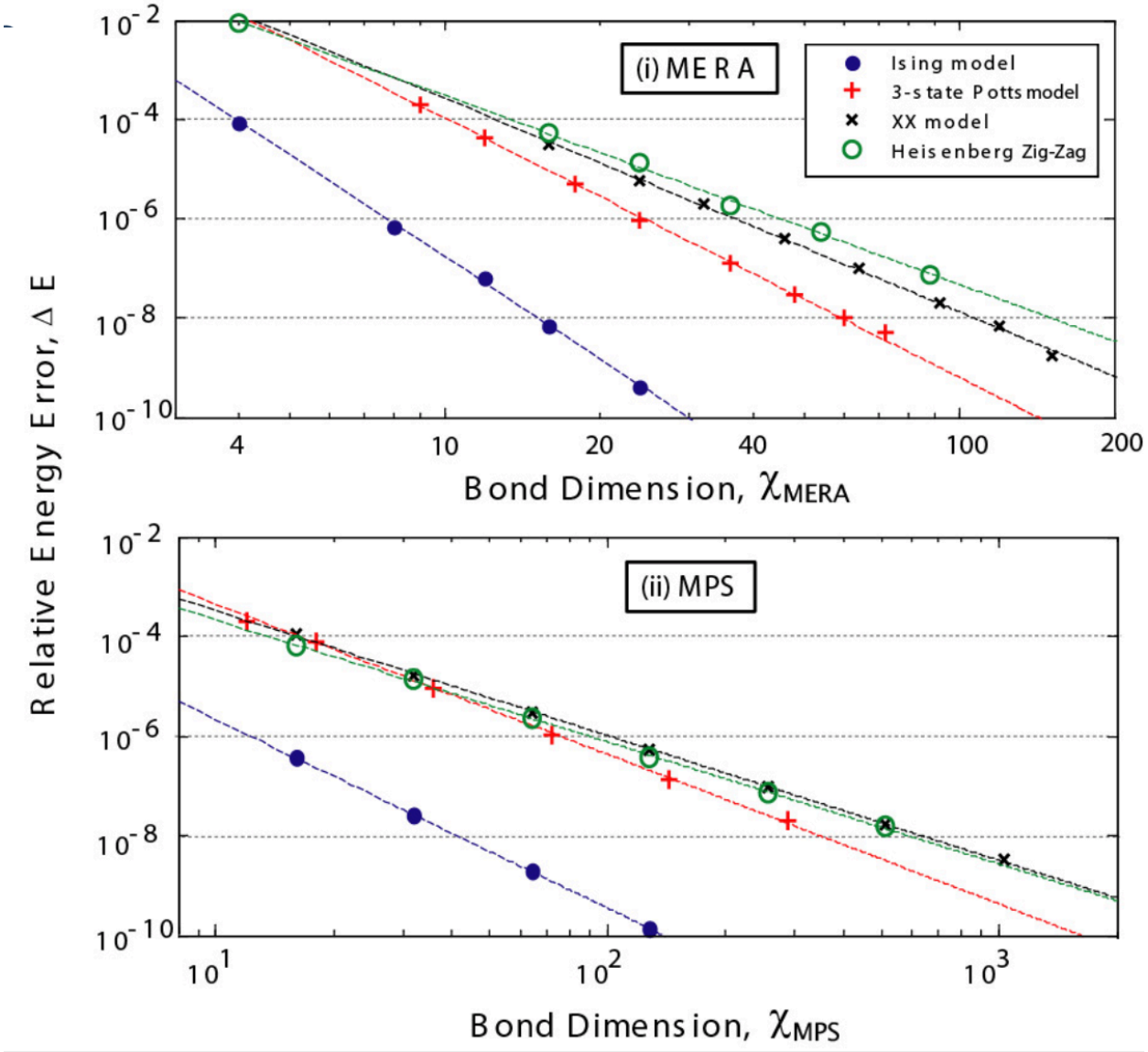
- 1D critical phases (gapless Hamiltonian)
- logarithmic scaling of entanglement entropy: $S \sim \log_3 N$
- polynomial decay of correlators

$$\langle O(0)O(r)\rangle \sim \frac{1}{r^{2\Delta}}$$

- not so easy to implement

Comparison of results

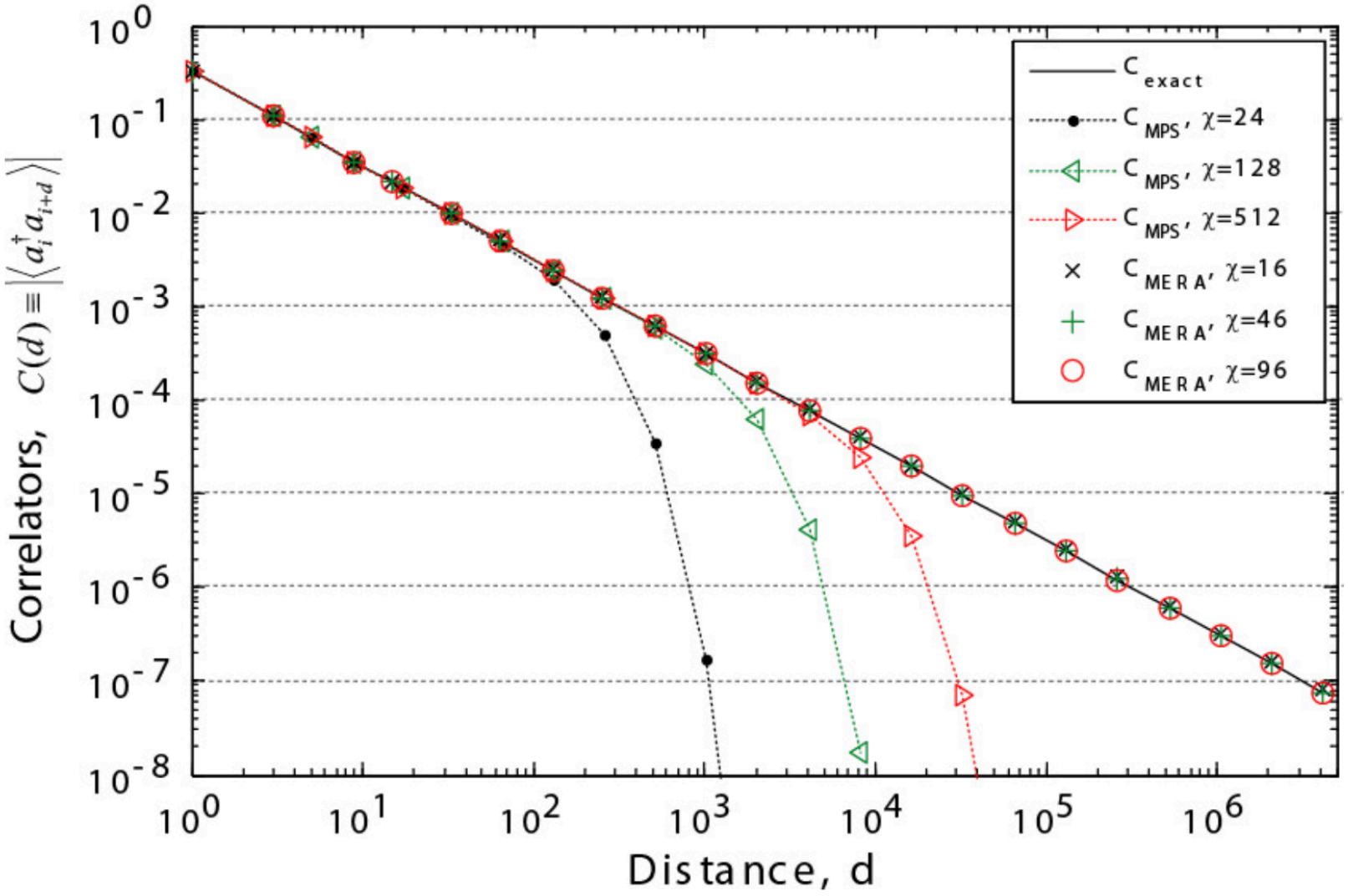
- ground state energies for critical spin chains



Comparison of results

- correlators for quantum XX model, critical with central charge $c = 1$

$$H_{XX} = \sum_j (\sigma_j^x \sigma_{j+1}^x + \sigma_j^y \sigma_{j+1}^y)$$

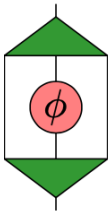


Conformal data from MERA

- two-point correlator of a critical ground state $\langle O(0)O(r)\rangle \sim \frac{1}{r^{2\Delta}}$

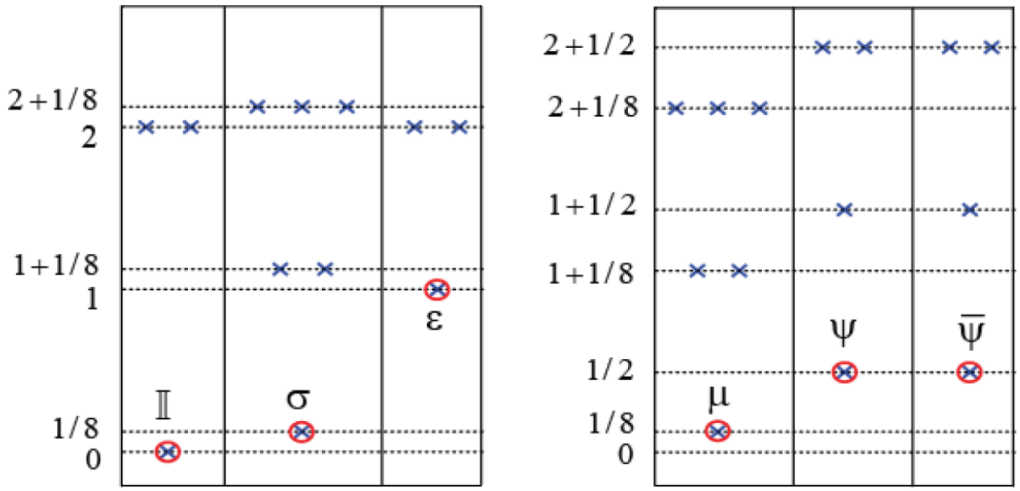
  Δ is known as the scaling dimension of the critical theory

- we can compute the scaling dimensions directly from the scale-invariant MERA

$\mathcal{S}(\phi) =$  $S(\phi) = \lambda\phi$ $\Delta = -\log_3\lambda$

- example: critical Ising model

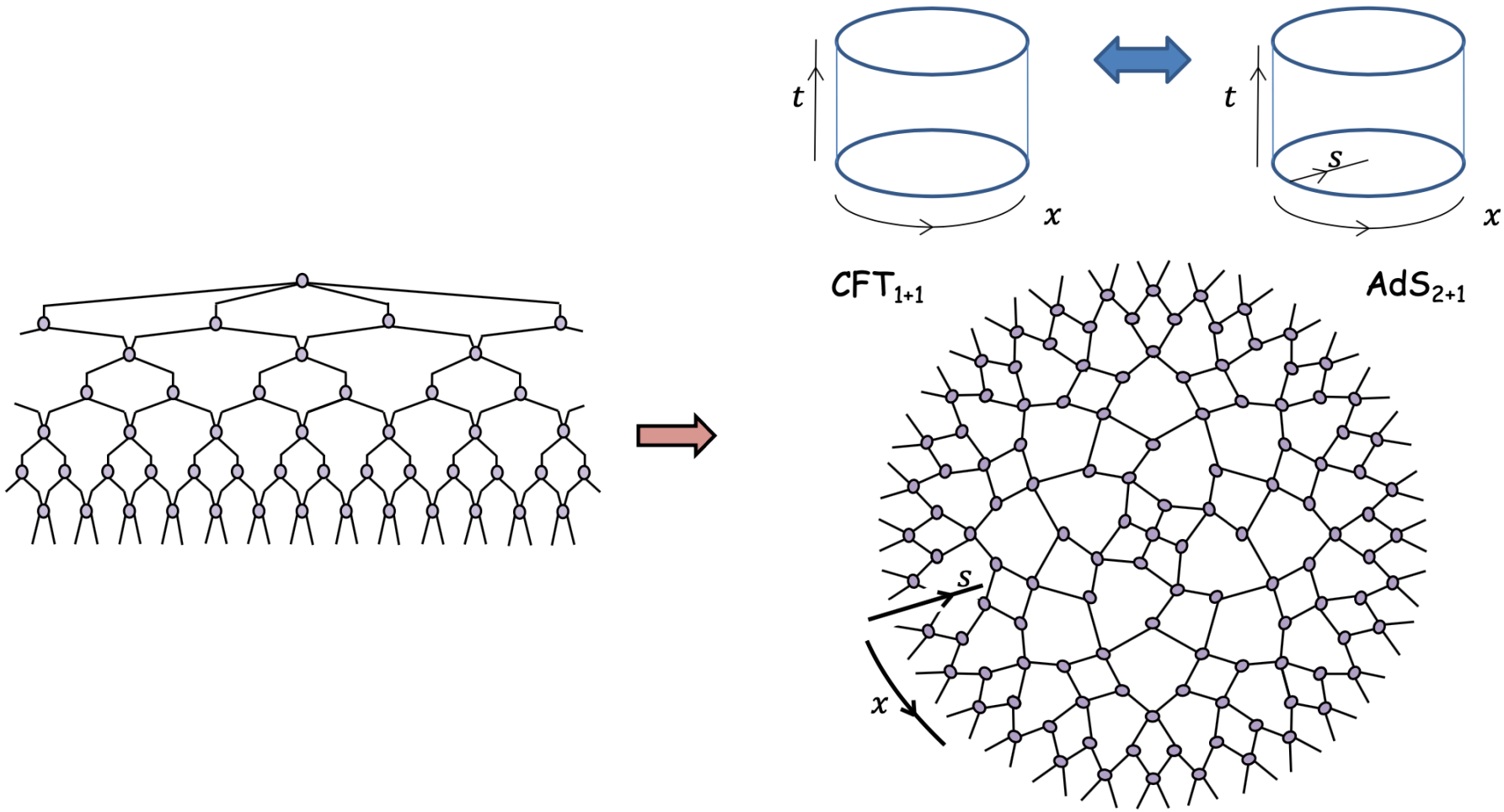
  scaling dimensions



primary fields	exact	MERA
spin	$\sigma = 0.125$	0.1250003
energy	$\varepsilon = 1$	1.0000200
disorder	$\mu = 0.125$	0.125002
fermion	$\psi = 0.5$	0.500004

MERA and holography

- it has been suggested that the MERA is a lattice realization of the AdS/CFT correspondence of string theory



B. Swingle, Phys. Rev. D 86, 065007 (2012)

Numerical Renormalization Group

- physical background

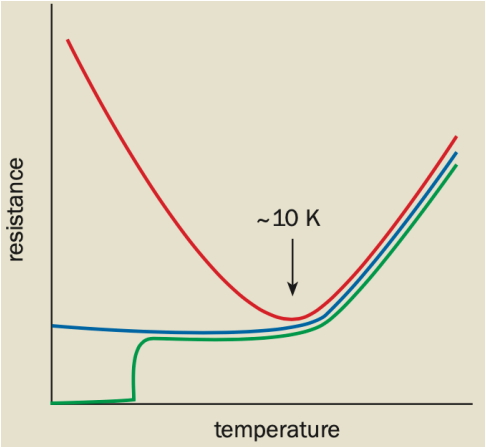
- Wilson's renormalization group method

Selected review articles:

L. Kouwenhoven and L. Glazman, Phys. World 14, 33 (2001)
K. G. Wilson, Rev. Mod. Phys. 47, 773 (1975)
R. Bulla, T. A. Costi, and T. Pruschke, Rev. Mod. Phys. 80, 395 (2008)

Electron transport at low temperatures

- electrical resistance



- normal metal

  drops with lowering temperature and saturates as the temperature is lowered below about 10 K due to static defects in the material

- superconductor

  below critical temperature, suddenly lose all their resistance to electrical current and become superconducting

- normal metal with magnetic impurities Kondo effect

  drops with temperature at first

  but increases logarithmically as the temperature is lowered further

The role of spin

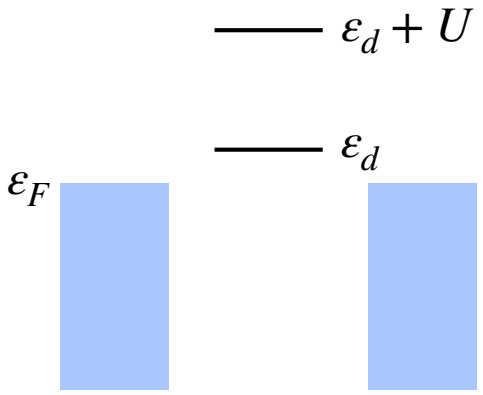
- Kondo effect only arises when the defects are magnetic

  i.e., when the total spin of all the electrons in the impurity atom is non-zero

- Anderson model - the simplest model of a magnetic impurity

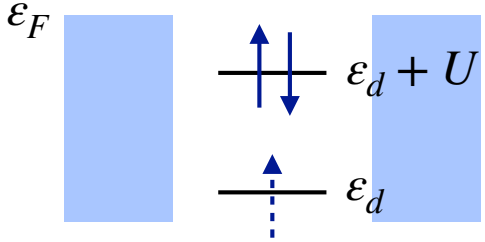
$$H_{AM} = \underbrace{\sum_{\sigma} \varepsilon_d \, n_{d\sigma} + U \, n_{d\uparrow} n_{d\downarrow}}_{H_{\text{imp}}} + \underbrace{\sum_{k\sigma} V_{kd}(c^{\dagger}_{k\sigma} d_{\sigma} + d^{\dagger}_{\sigma} c_{k\sigma})}_{H_{\text{int}}} + \underbrace{\sum_{k\sigma} \epsilon_k \, c^{\dagger}_{k\sigma} c_{k\sigma}}_{H_{\text{bath}}}$$

- Coulomb repulsion for two electrons occupying the same impurity site



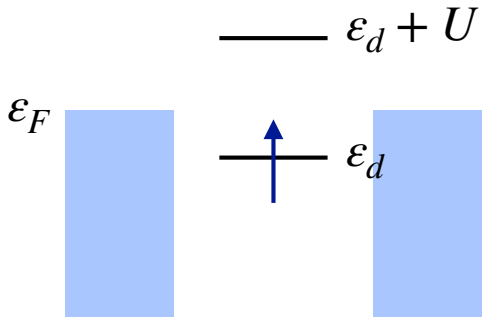
- $\varepsilon_d > \varepsilon_F$, $n_d = 0$

  no local magnetic moment



- $\varepsilon_d + U < \varepsilon_F$, $n_d = 2$

  no local magnetic moment



- $\varepsilon_d < \varepsilon_F$, $\varepsilon_d + U > \varepsilon_F$, $n_d = 1$

  local magnetic moment

Kondo model & Kondo formula

- at low T, resistivity is caused by the scattering between the conduction electrons and the impurity electron spin, not charge

- the interaction between the spin of the impurity and the conduction electrons is exchange interaction, which can be characterized by an exchange integral J

- Kondo model $$H_{KM} = \underbrace{-g\mu_B B S_z}_{H_{\text{imp}}} + \underbrace{J\vec{S}\cdot\vec{s}_0}_{H_{\text{int}}} + \underbrace{\sum_{k\sigma}\varepsilon_k c^{\dagger}_{k\sigma}c_{k\sigma}}_{H_{\text{bath}}}$$



Jun Kondo

- using perturbation theory, Kondo showed resistivity growth - Kondo formula

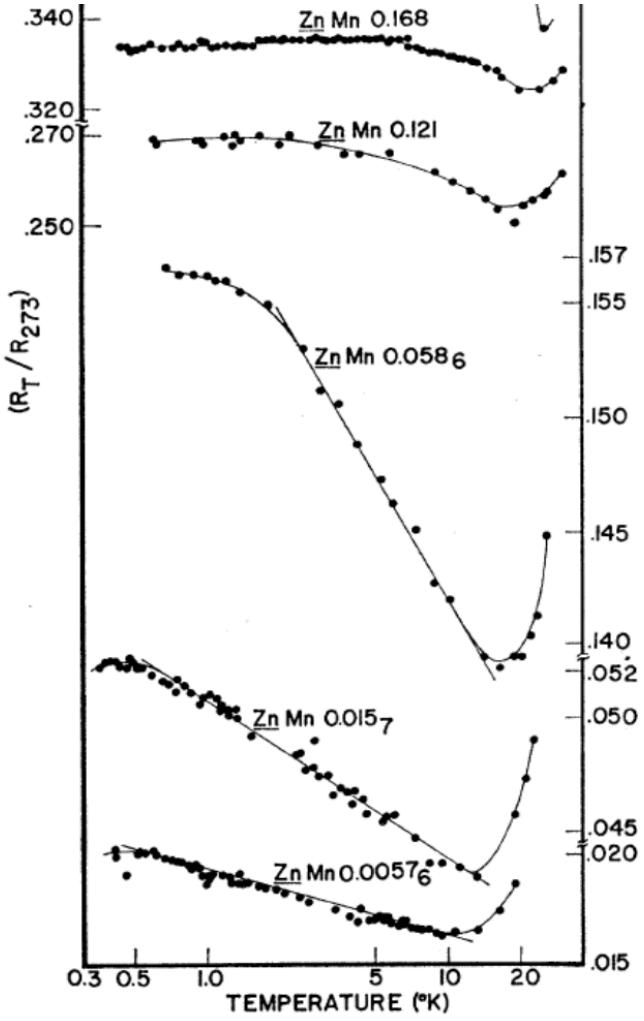
$R(T) = R_0\left(1 + 2JD\cdot\log\frac{k_B T}{\varepsilon_F}\right)$, D is the density of state assumed to be constant

- the dependence of the resistivity ρ on temperature T

$$\rho(T) = \rho_0 + aT^2 + c_m \ln\frac{\mu}{T} + bT^5$$

Kondo problem

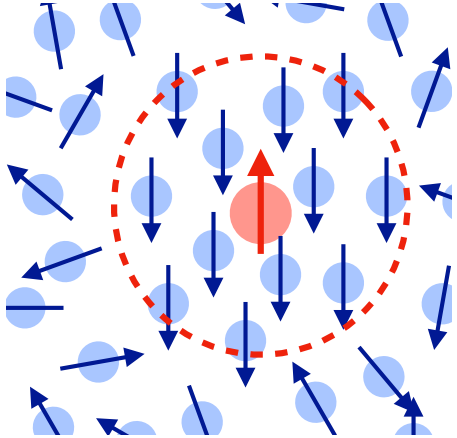
- Kondo resistivity formula: $\rho(T) = a - b \log T$
- when $T \to 0$, $\rho \to \infty$, the resistivity diverges as T approaches zero
- resistivity is logarithmic above T_K, but deviates from the log behavior at $T < T_K$
- at very low T the resistivity saturates at a certain value
- we need a non-perturbative theory to treat the Kondo effect at low $T \to$ Kondo problem
- intrinsically a strongly correlated quantum many body problem that cannot be solved by perturbing a Fermi liquid



Kondo screening cloud

- Anderson's poor man's scaling

  for $T < T_k$, the impurity spin and the conduction electron spins form a bound state with total spin $S = 0$



- the impurity spin is screened by conduction electrons

  won't be seen directly by other conduction electrons

Wilson's Numerical Renormalization Group (NRG)

- non-perturbative method

  provide an exact numerical solution

  Kondo problem is totally solved

- renormalization group theory for phase transitions

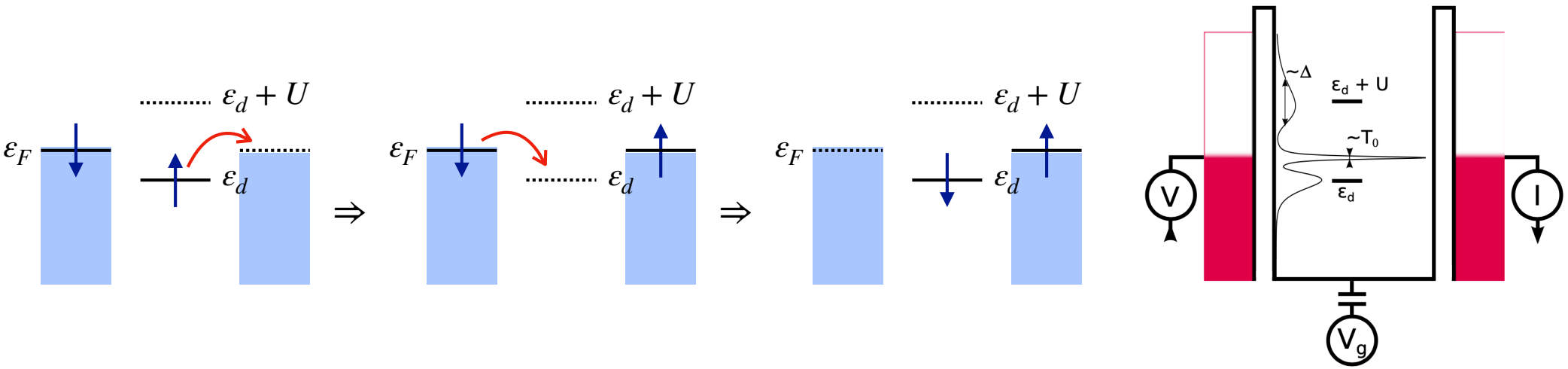
  1982 Nobel prize "for his theory for critical phenomena in connection with phase transitions"



Kenneth G. Wilson

Kondo resonance

initial state virtual state final state density of states



- effectively flip the spin of the impurity
- a pseudo energy level formed as a result of numerous spin-flip process
  an extra resonance at the Fermi energy → Kondo resonance
- since conductance is determined by electrons with energies close to the Fermi level, the extra resonance is very effective at scattering electrons
- the strong scattering contributes greatly to the resistance

Quantum impurity models

- Anderson impurity model

$$H_{AM} = \underbrace{\sum_{\sigma} \varepsilon_d \, n_{d\sigma} + U \, n_{d\uparrow} n_{d\downarrow}}_{H_{\text{imp}}} + \underbrace{\sum_{k\sigma} V_{kd}(c^{\dagger}_{k\sigma} d_{\sigma} + d^{\dagger}_{\sigma} c_{k\sigma})}_{H_{\text{int}}} + \underbrace{\sum_{k\sigma} \epsilon_k \, c^{\dagger}_{k\sigma} c_{k\sigma}}_{H_{\text{bath}}}$$

- Kondo impurity model

$$H_{KM} = \underbrace{-g\mu_B B S_z}_{H_{\text{imp}}} + \underbrace{J\vec{S} \cdot \vec{s}_0}_{H_{\text{int}}} + \underbrace{\sum_{k\sigma} \varepsilon_k c^{\dagger}_{k\sigma} c_{k\sigma}}_{H_{\text{bath}}}$$

- based on the Schrieffer-Wolff transformation, it was shown that Kondo model lies in the strong coupling regime of the Anderson impurity model

  J. R. Schrieffer and P. A. Wolff, Phys. Rev. 149, 491 (1966)

- the physics of both models will be the same at low temperatures

  one finds the correspondence $J = 2V^2(1/(U+\varepsilon_d) - 1/\varepsilon_d)$

Linear chain representation

- Anderson model hybridized with a local Wannier state $|0\sigma\rangle = f^\dagger_{0\sigma}|\mathrm{vac}\rangle$

$$V f^\dagger_{0\sigma} = \sum_k V_{kd} c^\dagger_{k\sigma}.$$

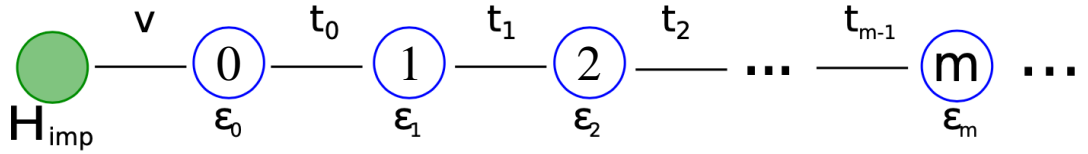
the value of V follows from the normalization $\{f_{0\sigma}, f^\dagger_{0\sigma}\} = 1$, $V = \left(\sum_k |V_{kd}|^2\right)^{1/2}$

- one can apply the Lanczos procedure for tridiagonalizing H_{bath} to obtain

$$H_{\mathrm{bath}} = \sum_{k\sigma} \varepsilon_k \, c^\dagger_{k\sigma} c_{k\sigma} \;\rightarrow\; \sum_{\sigma,n=0}^{\infty} \left[\epsilon_n \, f^\dagger_{n\sigma} f_{n\sigma} + t_n (f^\dagger_{n\sigma} f_{n+1\sigma} + H.c.)\right]$$

- we obtain a linear chain form of Anderson model

$$H_{AM} = \varepsilon_d \, n_d + U \, n_{d\uparrow} n_{d\downarrow} + V \sum_\sigma \left(f^\dagger_{0\sigma} d_\sigma + d^\dagger_\sigma f_{0\sigma}\right) + \sum_{\sigma,n=0}^{\infty} \left[\epsilon_n \, f^\dagger_{n\sigma} f_{n\sigma} + t_n (f^\dagger_{n\sigma} f_{n+1\sigma} + f^\dagger_{n+1\sigma} f_{n\sigma})\right]$$



- linear chain form of Kondo model

$$H_{KM} = -g\mu_B S_z + J\vec{S} \cdot \vec{s}_0 + \sum_{\sigma,n=0}^{\infty} \left[\epsilon_n f^\dagger_{n\sigma} f_{n\sigma} + t_n (f^\dagger_{n\sigma} f_{n+1\sigma} + f^\dagger_{n+1\sigma} f_{n\sigma})\right]$$

Separation of scales

- it is always possible to rewrite a quantum impurity model in the form of a semi-infinite linear chain

- truncating this chain to include m orbitals

$$H_m = \varepsilon_d n_d + U n_{d\uparrow} n_{d\downarrow} + V \sum_{\sigma} (f_{0\sigma}^{\dagger} d_{\sigma} + d_{\sigma}^{\dagger} f_{0\sigma}) + \sum_{\sigma,n=0}^{m} \epsilon_n f_{n\sigma}^{\dagger} f_{n\sigma} + \sum_{\sigma,n=0}^{m-1} t_n (f_{n\sigma}^{\dagger} f_{n+1\sigma} + f_{n+1\sigma}^{\dagger} f_{n\sigma})$$

- the truncated Hamiltonian H_m satisfying the recursion relation

$$H_{m+1} = H_m + \sum_{\sigma} \epsilon_{m+1} f_{m+1\sigma}^{\dagger} f_{m+1\sigma} + \sum_{\sigma} t_m (f_{m\sigma}^{\dagger} f_{m+1\sigma} + f_{m+1\sigma}^{\dagger} f_{m\sigma})$$

- with this recursion relation, one can iteratively diagonalize the Hamiltonian starting from H_0

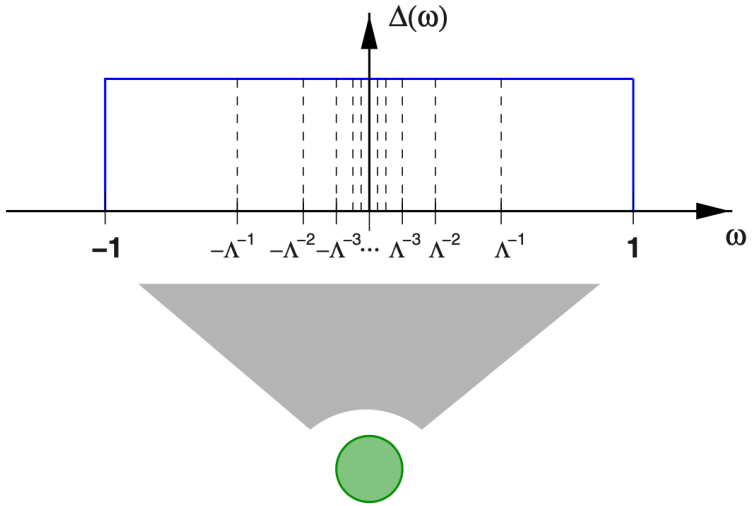
- at some point, we only able to retain the lowest many-body states of H_m

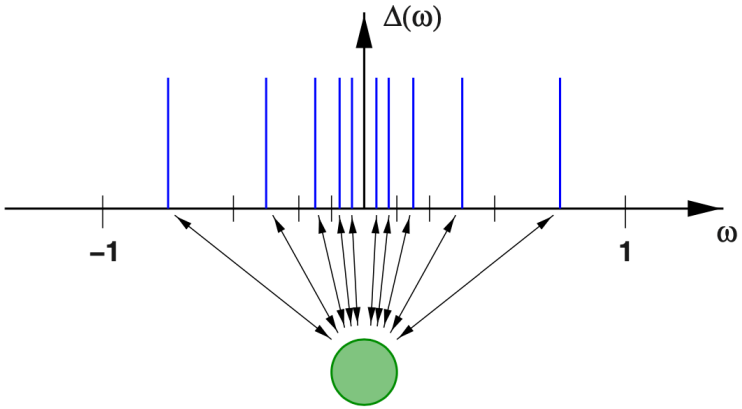
  we start neglecting some higher-energy states

  the validity of this procedure depends on whether the perturbation term is small

Logarithmic discretization approximation

- Wilson discretized the conduction band into positive and negative energy intervals $D_n^+ = [\Lambda^{-(n+1)}, \Lambda^{-n}], \qquad D_n^- = [-\Lambda^{-n}, -\Lambda^{-(n+1)}]$





- the quasi-continuous band is then approximated by a discrete one by keeping only a single conduction state from each interval

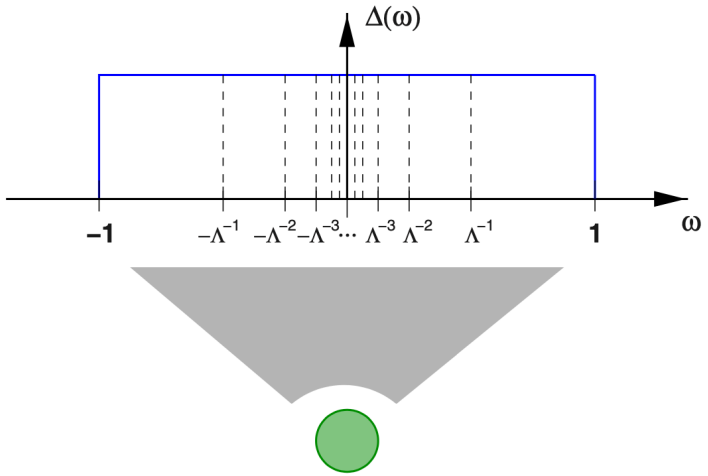
$$H_{\text{bath}} = \int_{-D}^{+D} d\epsilon\, \epsilon\, c_\epsilon^\dagger c_\epsilon \approx \sum_{n=0}^{\infty} \left(\varepsilon_{-n}\, c_{-n}^\dagger c_{-n} + \varepsilon_{+n}\, c_{+n}^\dagger c_{+n} \right)$$

$$\varepsilon_{\pm n} = \pm \frac{1}{2} D \left(\Lambda^{-n} + \Lambda^{-(n+1)} \right) = \pm \frac{1}{2} D \Lambda^{-n} \left(1 + \Lambda^{-1} \right) \qquad t_m \approx \tfrac{1}{2} D (1 + \Lambda^{-1}) \Lambda^{-m/2}$$

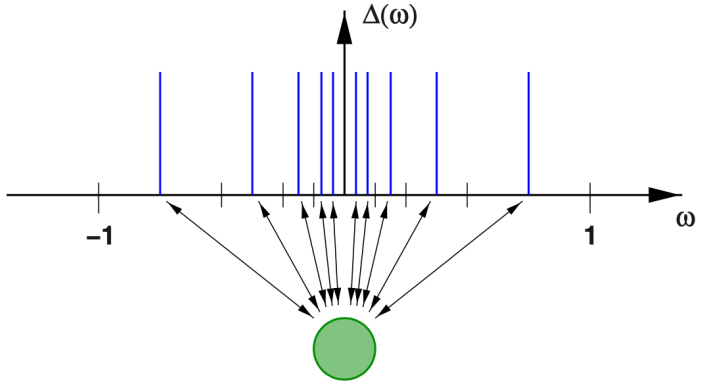
- hoppings and chemical potentials decaying sufficiently fast along the chain

General strategy of NRG

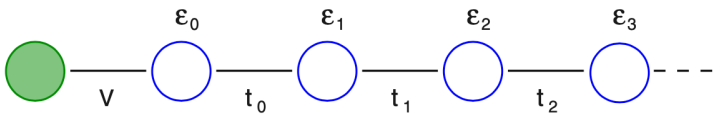
- division of the energy support of the bath spectral function into a set of logarithmic intervals



- reduction of the continuous spectrum to a discrete set of states logarithmic discretization



- mapping of the discretized model onto a semi-infinite chain
- iterative diagonalization of this chain



- further analysis static and dynamic properties of the quantum impurity model

Iterative diagonalization

- the chain Hamiltonian can be viewed as a series of Hamiltonian H_N which approaches H in the limit $N \longrightarrow \infty$

$$H = \lim_{N\to\infty} \Lambda^{-(N-1)/2} H_N$$

$$H_N = \Lambda^{(N-1)/2} \left[H_{\text{imp}} + \sqrt{\frac{\xi_0}{\pi}} \sum_\sigma (f_{0\sigma}^\dagger d_\sigma + d_\sigma^\dagger f_{0\sigma}) + \sum_{\sigma,n=0}^{N} \epsilon_n f_{n\sigma}^\dagger f_{n\sigma} + \sum_{\sigma,n=0}^{N-1} t_n (f_{n\sigma}^\dagger f_{n+1\sigma} + f_{n+1\sigma}^\dagger f_{n\sigma}) \right]$$

- the factor has been chosen to cancel the N dependence of the hopping matrix element between the last two sites of H

  useful for discussion of fixed points

- two successive Hamiltonians are related by

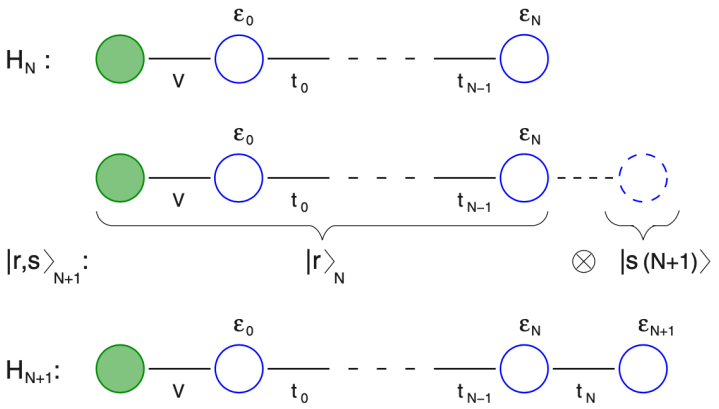
$$H_{N+1} = \sqrt{\Lambda} H_N + \Lambda^{N/2} \sum_\sigma \epsilon_{N+1} f_{N+1\sigma}^\dagger f_{N+1\sigma} + \Lambda^{N/2} \sum_\sigma t_N (f_{N\sigma}^\dagger f_{N+1\sigma} + f_{N+1\sigma}^\dagger f_{N\sigma})$$

- the starting point of the sequence of Hamiltonian is

$$H_0 = \Lambda^{-1/2} \left(H_{\text{imp}} + \sum_\sigma \epsilon_0 f_{0\sigma}^\dagger f_{0\sigma} + \sqrt{\frac{\xi_0}{\pi}} \sum_\sigma (f_{0\sigma}^\dagger d_\sigma + d_\sigma^\dagger f_{0\sigma}) \right)$$

Renormalization group transformation

- the recursion relation can now be understood in terms of a renormalization group transformation R, $H_{N+1} = R(H_N)$



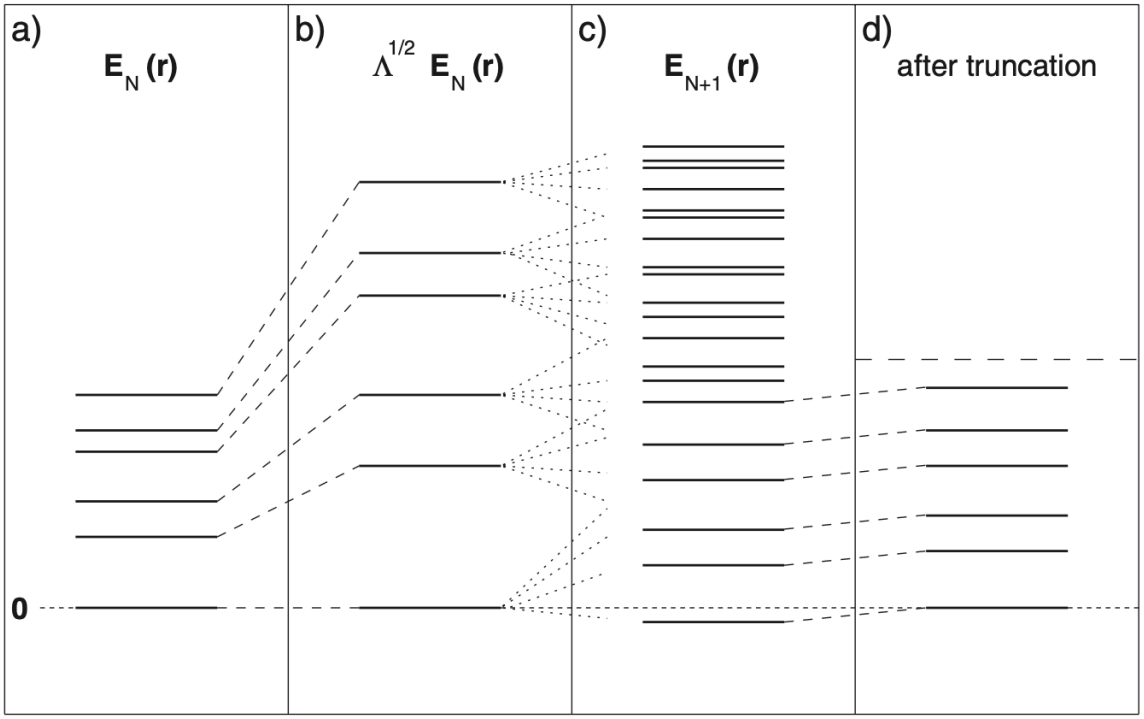
$$H_N|r\rangle_N = E_N(r)|r\rangle_N, \quad r = 1, \ldots, N_s$$

$$|r;s\rangle_{N+1} = |r\rangle_N \otimes |s(N+1)\rangle$$

$$H_{N+1} = \sqrt{\Lambda} H_N + \hat{X}_{N,N+1} + \hat{Y}_{N+1}$$

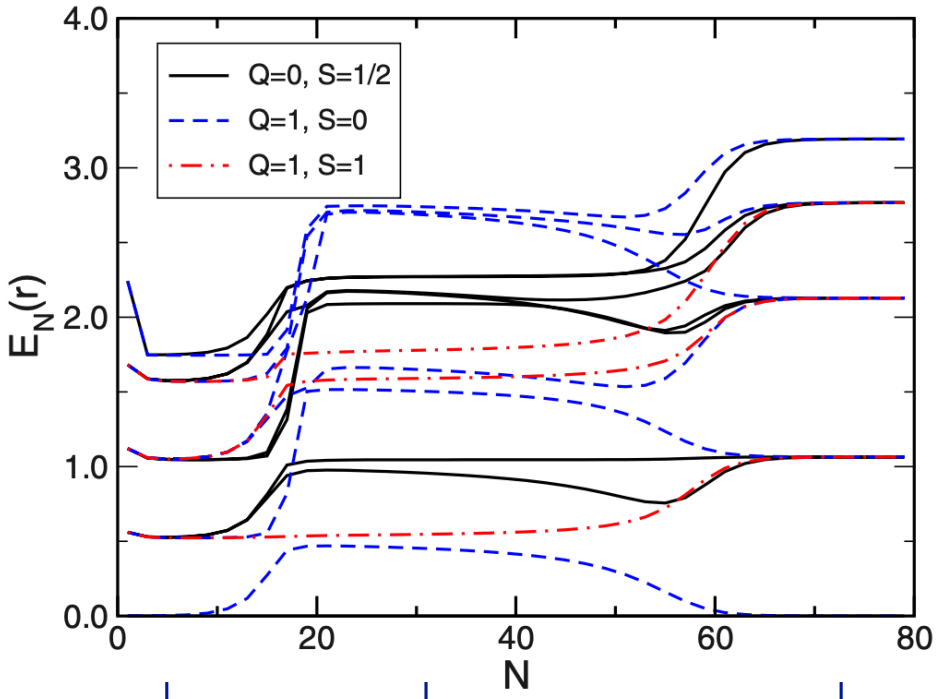
- many-particle spectrum of the Hamiltonian with the ground state energy set to zero

- after truncation, only the N_{keep} lowest-lying states are retained



Renormalization group flow

- the flow many-particle energies for the symmetric single-impurity Anderson model



labeled by total charge Q and total spin S

free-orbital fixed point

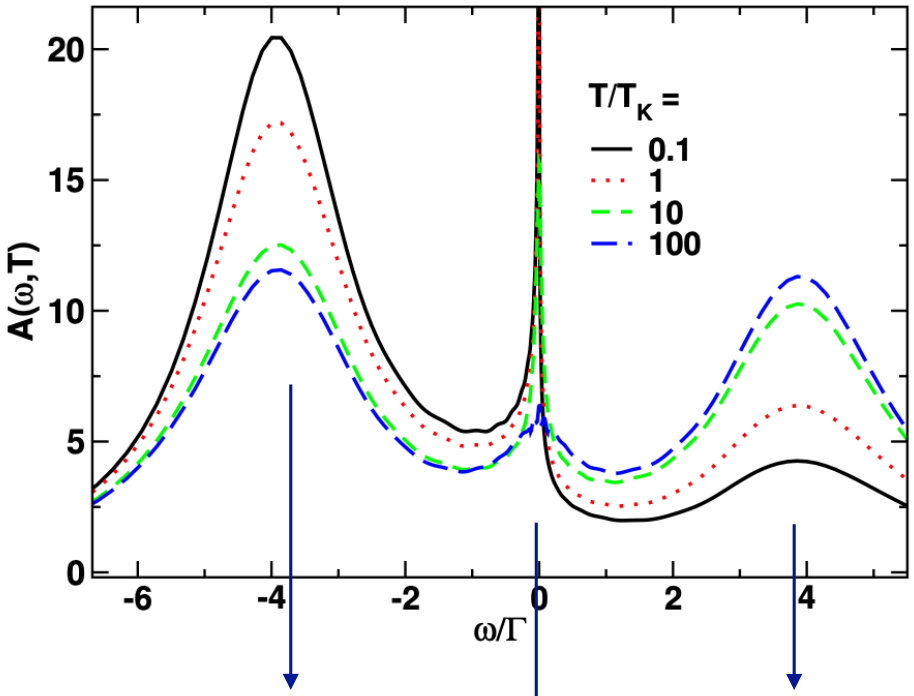
local-moment fixed point

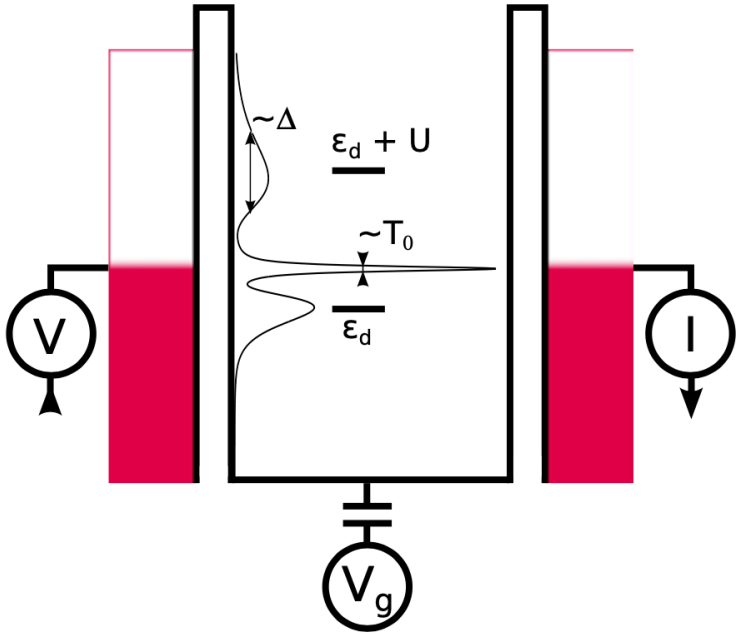
strong-coupling fixed point of a screened spin

Krishna-murthy, H. R., J. W. Wilkins, and K. G. Wilson, Phys. Rev. B 21, 1003 (1980)
Krishna-murthy, H. R., J. W. Wilkins, and K. G. Wilson, Phys. Rev. B 21, 1044 (1980)

Spectral function

- spectral function of Anderson model





S. Andergassen, T. A. Costi, and V. Zlatic, Phys. Rev. B 84, 241107(R) (2011)